Relativistic elastic shells: material support and cavity geometry

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Abstract: We derive the source moments that determine the leading tidal field and rotational clock shift inside a finite-thickness relativistic elastic shell. In weak gravity and slow rotation, these moments depend on the elastic response of the material and its two free vacuum faces. Two choices of elastic moduli, each causal near relaxation, give opposite signs for both observables. Both shells have oblate faces and gain rotational mass. The tide depends on differential flattening, bulk deformation, and rotational stress and energy; the central clock shift measures a distinct inverse-radius moment. A single constitutive action determines the material support and response. Nonnegative isotropic pressure cannot support a static spherical shell of positive-density matter with two free vacuum faces, whereas tangential elastic stress admits equilibria near relaxation. For sufficiently weak gravity these equilibria obey the dominant energy condition and have subluminal sound speeds. Their static cavities are locally flat, with clocks redshifted relative to infinity. At leading weak-gravity order, static redshift and linear frame dragging probe the same inverse-radius mass moment. Energy estimates establish radial linear stability and exclude exponentially growing odd-parity modes for weakly gravitating spherical equilibria. Nonlinear compression can nevertheless produce unbounded gradients while the material remains causal. Nonspherical even-parity and rotating stability, together with quantitative bounds at finite rotation, remain open. The cavity observables distinguish material responses that surface shape and total mass alone do not determine.

Keywords: relativistic elasticity, self-gravitating shells, frame dragging, tidal fields

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1 Introduction

Inside a static spherical shell, a clock can remain at rest without support while ticking more slowly than a clock at infinity. The regular vacuum cavity is locally flat; the redshift compares clocks in different parts of the same spacetime. Slow rotation adds a relative rotation of the cavity's inertial frame, followed at second order by tidal curvature that changes the separation of freely falling tracers. We ask how the matter surrounding the cavity determines these measurements.

The support problem comes first. Gravity pulls the shell inward, but neither vacuum face exerts a force on the material. A static spherical shell cannot meet these conditions with nonnegative isotropic pressure alone. We use a finite-thickness elastic annulus, whose tangential compression supplies the missing support. One elastic action determines both the equilibrium stress and its response to rotation. The cavity contains only test clocks and freely falling tracers; their stress-energy is neglected.

Our main result is an explicit relation between the elastic response and two cavity observables: the leading tide and the rotational correction to the central clock rate. Their source moments probe different features of the material. In weak gravity, the static redshift and linear frame dragging weight each spherical mass layer by its inverse radius. The leading cavity tide is more sensitive to matter near the inner face. It depends on how the two faces flatten relative to one another, as well as on deformation and stress within the material. Two choices of elastic moduli, each causal near relaxation, give oblate faces, flattened along the rotation axis, yet interchange the stretching and focusing directions of nearby tracers. The shape of either surface alone therefore does not determine the cavity geometry. Rotation also changes the central clock rate. Its correction depends on another source moment, so an increase in total mass need not make that clock run more slowly.

Finite-thickness elastic shells and their two free boundaries were studied by Losert-Valiente Kroon [1], including a Newtonian existence argument. Brito, Carot and Vaz [2] constructed relativistic elastic thick shells with causal propagation and dominant-energy bounds, matched to cosmological regions. Here both adjacent regions are vacuum, and the exterior is asymptotically flat. Einstein–Vlasov shells can also have vacuum centers [3], and Einstein clusters provide a circular-orbit support mechanism [4]. We use the material framework of relativistic elasticity [5, 6] and established static and rotating elastic-body theory [7, 8]. The new calculation identifies the source moments for the cavity tide and clock shift and evaluates them for two materials with opposite responses, both causal near relaxation. Static shell existence and the rotating elastic-body framework provide the foundations; the rotating existence result is an application of that established theory.

The connection to subluminal warp shells is the geometry of a moving body with a vacuum interior [9]. Recent analyses emphasize the distinction between coordinate

motion and material motion, and between a prescribed stress tensor and a matter model with its own equations of motion [10, 11]. Here the action specifies those equations and the stresses at both faces. The solutions describe gravitating elastic bodies with cavities. Their nonconstant lapse and curved spatial slices lie outside the unit-lapse, spatially flat ansatz of many warp-metric restrictions [12, 13]; Olum's superluminal-travel criterion [14] has different hypotheses. Neither propulsion nor a travel-time advantage follows from the cavity.

Sections 2–5 develop this sequence: material support, static redshift, frame dragging, and the changes in curvature and clock rate caused by rotation. Section 6 separates linear stability from nonlinear smooth evolution and discusses uniform motion and additional loads. The appendices give the coefficient derivations, dynamical proofs, and exact transformation to a uniformly moving shell.

The signature is $(-+++)$, $R^a{}_{bcd} = \partial_c\Gamma^a_{db} - \partial_d\Gamma^a_{cb} + \Gamma^a_{ce}\Gamma^e_{db} - \Gamma^a_{de}\Gamma^e_{cb}$, and the cosmological constant vanishes. We use $c = 1$ and usually $G = 1$, restoring G in existence and dynamical arguments. The mass function m, total mass M, angular momentum J, and four-momentum P^μ are geometrized; when G is explicit, their physical values are m/G, M/G, J/G, and P^μ/G.

2 Free boundaries and material support

Let r be the areal radius, so that each sphere has area $4\pi r^2$. Write the static spherical metric as

$$ds^2 = -e^{2\Phi(r)}dt^2 + \frac{dr^2}{1-2m(r)/r} + r^2 d\Omega^2, \qquad F(r) = 1 - \frac{2m(r)}{r} > 0. \tag{2.1}$$

For $T^{\hat{a}}{}_{\hat{b}} = \mathrm{diag}(-\rho, p_r, p_t, p_t)$, the independent equations can be written

$$m' = 4\pi r^2\rho, \qquad \Phi' = \frac{m + 4\pi r^3 p_r}{r(r-2m)}, \tag{2.2}$$

$$p_r' = -(\rho + p_r)\Phi' + \frac{2(p_t - p_r)}{r}. \tag{2.3}$$

Here ρ is the energy density, p_r the radial pressure, and p_t the tangential pressure, all measured in the material rest frame. The function $m(r)$ is the enclosed geometrized mass; e^Φ sets the rate of a stationary clock relative to the time coordinate t. The last equation is radial stress conservation. Together with the first two, it supplies the angular Einstein equation in the smooth bulk. For a shell occupying $a \le r \le b$, with an empty regular cavity, $m(a) = 0$. Matching to vacuum without a surface layer requires $p_r(a) = p_r(b) = 0$, with continuous induced metric and extrinsic curvature on each timelike boundary [15]. The induced metric gives distances and times along the boundary; the extrinsic curvature describes how it is embedded in spacetime. Continuity of both is called Darmois matching and excludes an additional sheet of

stress-energy on either face. For the metric above, the nontrivial mixed extrinsic curvatures of a static face, with the same increasing-r normal on its two sides, are $\mathcal{K}^t{}_t = \sqrt{F}\,\Phi'$ and $\mathcal{K}^\theta{}_\theta = \mathcal{K}^\phi{}_\phi = \sqrt{F}/r$. Continuity of m and e^Φ, together with $p_r = 0$, makes these equal to their vacuum values. The density itself may jump: it enters m', but does not create a surface layer when these matching conditions hold.

The support mechanism is visible directly in stress conservation:

$$-p_r' - (\rho + p_r)\Phi' + \frac{2(p_t - p_r)}{r} = 0, \qquad \int_a^b \frac{2(p_t - p_r)}{r}\,dr = \int_a^b (\rho + p_r)\Phi'\,dr. \tag{2.4}$$

Multiplying the local balance by $\sqrt{F}$ gives proper radial force densities. The integral identity follows from $p_r(a) = p_r(b) = 0$. If the gravitational term on the right is positive, tangential pressure must exceed radial pressure somewhere. Here positive p_i denotes compression. Its anisotropic contribution can point outward even though gravity points inward; a traction-free face requires $p_r = 0$, not $p_t = 0$.

Proposition 2.1 (Isotropic inner-boundary obstruction)**.** *Let $0 < a < b$, and let a horizon-free static spherical shell have continuous, nonnegative density and pressure, isotropic stress, and nonzero density somewhere in (a, b). Assume the bulk Einstein equations, differentiability there, $m(a) = 0$, and vacuum matching without a surface layer at a. These assumptions are incompatible.*

Proof. Equations (2.2)–(2.3) give $m \geq 0$ and

$$p' = -\frac{(\rho + p)(m + 4\pi r^3 p)}{r(r - 2m)} \leq 0. \tag{2.5}$$

Since $p(a) = 0$ and $p \geq 0$, monotonicity forces $p = 0$ throughout the shell. Continuity and nontriviality of ρ provide an interior point with both $\rho > 0$ and $m > 0$. Equation (2.5) then gives $p' < 0$, contradicting $p = 0$. □

Tangential stress can supply the required support while obeying the dominant energy condition; Appendix A gives an explicit example with $p_r = 0$. To determine how a shell responds when its load changes, we need a law relating stress to deformation.

2.1 A material law for support and response

Material coordinates X^A are labels carried by the particles. A fixed Euclidean material metric k_{AB} records their separations in the relaxed, unstressed body. The spacetime metric gives their actual proper separations. In the three principal directions, let n_i^{-1} be the stretch relative to relaxation. Thus $n_i = 1$ at relaxation and compression increases n_i; the product $n = n_1 n_2 n_3$ is the relative particle number density.

These quantities have a covariant definition. Set $H^{AB} = g^{ab}\partial_a X^A \partial_b X^B$; the n_i^2 are the eigenvalues of $H^{AC}k_{CB}$. The material map has rank three and a timelike kernel, spanned by the future unit material velocity u^a, so $u^a\partial_a X^A = 0$. We choose a compressional energy and the quasi-Hookean shear invariant of Ref. [5]:

$$s^2 = \frac{1}{12}\sum_{i<j}\left(\frac{n_i}{n_j} - \frac{n_j}{n_i}\right)^2,$$
$$\rho(n_i) = m_0 n + \frac{\kappa}{2}(n-1)^2 + \mu_0 n s^2, \qquad p_i = n_i\frac{\partial\rho}{\partial n_i} - \rho. \tag{2.6}$$

The three terms in ρ account for rest mass, volume change, and shear. The constant m_0 is the relaxed energy density, while κ and μ_0 are the bulk and shear moduli at relaxation. All three have units of energy density; away from relaxation the shear coefficient is $\mu_0 n$. The material action is $S_m = -\int_{\mathcal{B}} \rho\sqrt{-g}\, d^4x$ over the occupied material worldtube $\mathcal{B}$; its exterior is vacuum. Its variations give

$$T_{ab} = 2\rho_{H^{AB}}\partial_a X^A \partial_b X^B - \rho g_{ab}, \qquad \nabla_a\left(2\rho_{H^{AB}}\nabla^a X^B\right) = 0 \tag{2.7}$$

in Cartesian material coordinates, together with zero physical traction on each free face. Since $u^a\partial_a X^A = 0$, the tensor obeys $T_{ab}u^b = -\rho u_a$; its spatial eigenvalues are the p_i in Eq. (2.6). The law is used only near relaxation. It describes a homogeneous, isotropic, perfectly elastic solid: there is no plasticity, viscosity, or fracture criterion. Material isotropy means that the relaxed body has no preferred local direction. A deformation can still produce unequal principal pressures, including the difference between p_r and p_t needed for shell support. There is no separate surface energy or surface tension in this action. The solid can have nonzero density at a free face. In the distributional conservation law, extending its stress by zero into vacuum produces a surface term proportional to $T^{ab}n_a$; zero traction removes that term. Thus a sharp material boundary is consistent with stress conservation, without prescribing an additional transition layer. The state $n_i = 1$ has zero stress. Its transverse and longitudinal squared characteristic speeds are respectively μ_0/m_0 and $(\kappa+4\mu_0/3)/m_0$. Transverse waves probe shear, while longitudinal waves also compress the material. Positive shear stiffness allows neighboring radial layers to support one another while both faces remain free.

The stress tensor has a material rest frame and three principal stresses, so it is of Hawking–Ellis Type I [16]. Its rest-frame energy density is ρ, and its principal pressures are p_1, p_2, p_3. The null (NEC), weak (WEC), and dominant (DEC) energy conditions therefore reduce to

$$\begin{aligned} \text{NEC} &\iff \rho + p_i \geq 0 \quad \text{for each } i, \\ \text{WEC} &\iff \rho \geq 0 \text{ and NEC}, \\ \text{DEC} &\iff \rho \geq |p_i| \quad \text{for each } i. \end{aligned} \tag{2.8}$$

These rest-frame inequalities characterize the energy conditions for all causal observers. In particular, the DEC requires nonnegative energy density and a causal energy flux for every timelike observer.

3 Elastic equilibrium, mass, and cavity redshift

We distinguish material radii $A \le R \le B$ in the relaxed reference annulus from physical areal radii $a \le r \le b$ in the equilibrium shell. The strain determines the map between them. The dimensionless static parameters are

$$\beta = B/A, \qquad \chi = Gm_0A^2, \qquad k = \kappa/m_0, \qquad \mu = \mu_0/m_0. \tag{3.1}$$

At fixed (β, k, μ), weak gravity means $\chi \to 0$. Rotation introduces $\varepsilon_\Omega = (A\varpi)^2$, where ϖ is the angular velocity measured relative to time at infinity. These parameters separate geometry, gravitational strength, and material stiffness; the individual choices of length and density units do not select a special shell. The examples below use $(\beta, k, \mu) = (2, 1/5, 1/10)$. Their relaxed sound speeds are $c_T = 1/\sqrt{10}$ and $c_L = 1/\sqrt{3}$, so this is a relativistically stiff ideal material. Weak gravity at fixed (k, μ) is therefore not a nonrelativistic material limit. In particular, the leading rotational calculation retains special-relativistic strain and stress contributions even as $\chi \to 0$. All small-parameter limits below keep (β, k, μ) fixed, with positive moduli. Slow rotation must also keep the strain small. Softer materials deform more at a given rim speed $B|\varpi|$, so a small speed alone is not an error bound for the deformation expansion.

For a spherical material shell $A \le R \le B$, put $\nu = n_r = \sqrt{F}\, dR/dr$ and $\eta = n_t = R/r$. In material radius the full static equations are

$$\begin{aligned} r_R &= \frac{\sqrt{F}}{\nu}, \qquad & m_R &= 4\pi r^2 \rho\, r_R, \\ \eta_R &= \frac{1}{r} - \frac{\eta}{r} r_R, \quad & \Phi_R &= \frac{m + 4\pi r^3 p_r}{r(r-2m)} r_R, \\ \nu_R &= \frac{[-(\rho + p_r)\Phi_r + 2(p_t - p_r)/r] r_R - (\partial_\eta p_r)\eta_R}{\partial_\nu p_r}. \end{aligned} \tag{3.2}$$

Here p_r, p_t are evaluated from Eq. (2.6) at $(n_1, n_2, n_3) = (\nu, \eta, \eta)$. The boundary conditions are

$$m(A) = 0, \quad p_r(A) = p_r(B) = 0, \quad \Phi(B) = \tfrac{1}{2}\log[1 - 2m(B)/r(B)]. \tag{3.3}$$

The material strain determines the physical radii.

Theorem 3.1 (Nonlinear spherical elastic shells)**.** *Fix $0 < A < B$ and positive m_0, κ, μ_0 with $\kappa + 4\mu_0/3 < m_0$. For all sufficiently small $G > 0$, the material law (2.6) admits a locally unique spherical static shell near $r = R$, with a flat vacuum cavity, a Schwarzschild exterior, and zero traction at both faces. The material satisfies strict DEC and has real, strictly subluminal characteristic speeds.*

Proof. Restore G by replacing $4\pi\rho$ and $4\pi p_r$ in the mass and lapse equations by $4\pi G\rho$ and $4\pi G p_r$. Choose the inner physical radius $r(A) = q$ and integrate outward from $m(A) = 0$. Since $\partial_\nu p_r = \kappa + 4\mu_0/3 > 0$ at relaxation, the inner traction condition determines $\nu(A)$ smoothly from q near A. The bulk equations are smooth on a neighborhood of the relaxed solution throughout $[A, B]$. Let $\mathcal{R}(q, G)$ be the resulting outer traction; the remaining boundary condition is $\mathcal{R}(q, G) = 0$.

At $G = 0$, the linear displacement $u = \partial r/\partial q$ solves $u'' + 2u'/R - 2u/R^2 = 0$, $u(A) = 1$, with zero inner traction. Hence

$$u(R) = \frac{4\mu_0 R/A + 3\kappa A^2/R^2}{4\mu_0 + 3\kappa}, \qquad \partial_q \mathcal{R}(A, 0) = -\frac{12\kappa\mu_0}{A(4\mu_0 + 3\kappa)}\left(1 - \frac{A^3}{B^3}\right) < 0. \tag{3.4}$$

The implicit function theorem gives $q(G)$ and local uniqueness in the spherical class; exterior normalization fixes the lapse constant. For small positive G, the density remains positive, $F > 0$, and the material map remains invertible. The free-traction conditions give Darmois matching. Strict DEC and the characteristic gaps hold at relaxation and persist uniformly on the compact material interval and sphere of wave directions. More explicitly, the kinetic acoustic matrix is positive at relaxation, and the symmetric spatial acoustic matrix lies strictly between zero and the kinetic matrix for every unit wave covector. These strict matrix inequalities persist under small strain, including at repeated principal densities; no differentiable choice of eigenvectors is needed. □

This spherical construction complements the Newtonian shell construction [1] and general relativistic static-body theory [7]. Its small-coupling threshold is not quantified. The zero-shear limit is singular for this construction: the shooting derivative in Eq. (3.4) vanishes as $\mu_0 \to 0$. The theorem therefore cannot be extended to a perfect fluid by setting the shear modulus to zero.

3.1 A weakly gravitating example

For $A = 10$, $B = 20$, $m_0 = 10^{-6}$, $\kappa = m_0/5$, and $\mu_0 = m_0/10$, with $G = 1$ in a chosen geometrized length unit, shooting gives

$$r(A) \simeq 9.98529, \quad r(B) \simeq 19.98318, \quad M \simeq 0.0292968. \tag{3.5}$$

The computed profile has $n_r, n_t \in [0.99, 1.01]$, the constitutive domain treated in Appendix B. Its smallest computed DEC margin $\rho - \max(|p_r|, |p_t|)$ is about $1.0007 m_0$. The decimal values illustrate the equilibrium; the existence proof above is independent of this calculation. Figure 1 shows how this shell is supported. Near the inner face, radial pressure rises outward, so its gradient force points inward. Tangential compression supplies the outward term in Eq. (2.4); farther out, the pressure gradient reverses and also helps oppose gravity. Stress depends on all three strains: radial

extension can coexist with positive radial pressure because the tangential directions are compressed.

The density has a finite jump to vacuum at each free surface. The traction conditions give Darmois matching without a surface stress; this elastic example is piecewise smooth. The prescribed density in Appendix A instead vanishes smoothly at the faces.

Appendix B derives the acoustic matrices from the action and proves the uniform bounds

$$\frac{1}{16} \le c_{\mathrm{sound}}^2 \le \frac{2}{5}, \qquad \rho - \max_i |p_i| > \frac{24}{25} m_0, \qquad n_i \in [0.99, 1.01], \tag{3.6}$$

for the selected material ratios. The sound speeds are measured in the local material rest frame. These conservative bounds hold for every state in the displayed strain range. They apply to an equilibrium as long as its strain remains in that range. They constrain local signal propagation; the growth of perturbations of the whole gravitating shell is a separate question, treated in Section 6.

To give the example a physical scale, choose a length unit L_0. A quoted mass $\widehat{M}$ then corresponds to $M_{\mathrm{phys}} = (c^2 L_0/G_N)\widehat{M}$; energy density and stress scale as $c^4/(G_N L_0^2)$, and angular frequency as c/L_0. Weak compactness alone does not fix a laboratory mass or stress scale. The chosen elastic action has not been identified with a laboratory material.

3.2 The cavity clock and the mass at infinity

With the equilibrium fixed, the lapse determines the clock comparison. We keep G explicit in this subsection and write $e^{2\Lambda} = F^{-1}$. In the regular cavity, $m = 0$ and $\Phi = \Phi_c$ is constant. Rescaling time there by $T = e^{\Phi_c} t$ gives the Minkowski metric. The rate relative to a clock at infinity is fixed by matching through the shell:

$$\alpha_c = e^{\Phi_c} = \sqrt{1 - 2M/b}\, \exp\left[- \int_a^b \frac{m + 4\pi G r^3 p_r}{r(r - 2m)}\, dr \right]. \tag{3.7}$$

For the numerical example, $\alpha_c \simeq 0.99811$. A stationary cavity clock has $d\tau = \alpha_c\, dt$, and light of locally measured frequency ν_c received by an observer at rest at infinity has $\nu_\infty = \alpha_c \nu_c$, because the photon energy associated with the stationary time symmetry is conserved along the ray. The static observer's proper acceleration is $\sqrt{F}\,\Phi'$ and vanishes throughout the cavity. A local experiment can detect no static tidal field in this cavity, whereas comparison with a distant clock detects the redshift. This is a property of the isolated shell; an imposed external field would require a different solution.

The pressure that supports the shell also contributes to the mass measured at infinity. To relate that mass to the material energy, the spherical field equations give the exact identity

$$\left(r^2 e^{\Phi - \Lambda} \Phi'\right)' = 4\pi G r^2 e^{\Phi + \Lambda} (\rho + p_r + 2p_t). \tag{3.8}$$

It follows by adding the orthonormal time and spatial Einstein equations, or by differentiating the left side and substituting Eq. (2.2) and stress conservation. Its inner value vanishes and its outer value is M. Therefore

$$\frac{M}{G} = 4\pi \int_a^b e^{\Phi+\Lambda}(\rho + p_r + 2p_t) r^2 \, dr. \tag{3.9}$$

This is the Komar mass, defined by the stationary time symmetry. It equals the Arnowitt–Deser–Misner (ADM) mass at spatial infinity. In equilibrium it also equals $M/G = 4\pi \int_a^b \rho r^2 dr$. The pressure and redshift factors in the Komar integral therefore cannot be varied independently.

The proper energy $\mathcal{M}_{\rm prop} = 4\pi \int_a^b \rho r^2 dr/\sqrt{F}$ instead sums the energies measured in local material rest frames. Write $\epsilon = \rho/n$ for energy per relaxed material volume. Since $n = \sqrt{F}\, R^2/(r^2 r_R)$, the same integrals become

$$\frac{M}{G} = 4\pi \int_A^B R^2 \epsilon \sqrt{F}\, dR, \qquad \mathcal{M}_{\rm prop} = 4\pi \int_A^B R^2 \epsilon \, dR. \tag{3.10}$$

Thus $\mathcal{M}_{\rm prop} - M/G > 0$ for a nontrivial positive-density static shell. The ADM mass already includes gravitational binding. The reference rest energy $\mathcal{M}_0 = 4\pi m_0 (B^3 - A^3)/3$ is fixed by material content; $\mathcal{M}_{\rm prop}$ also includes strain energy.

On the weak-coupling branch, strain is $O(G)$ and the relaxed specific energy is stationary, so $\epsilon = m_0 + O(G^2)$. With $\mathfrak{m}_0(R) = 4\pi m_0 (R^3 - A^3)/3$, expanding Eq. (3.10) gives

$$\begin{aligned} \frac{M}{G} &= \mathcal{M}_0 - G\mathcal{B}_0 + O(G^2), \\ \mathcal{B}_0 &= 4\pi m_0 \int_A^B R \mathfrak{m}_0(R)\, dR = \frac{16\pi^2 m_0^2}{3} \left[\frac{B^5 - A^5}{5} - \frac{A^3 (B^2 - A^2)}{2} \right] > 0. \end{aligned} \tag{3.11}$$

The coefficient is the Newtonian binding energy of the relaxed annulus. The error statement is at fixed annulus and moduli; it gives no bound at arbitrary compactness.

4 Rotation and inertial-frame dragging

Rotation changes the comparison between the cavity's inertial axes and those at infinity. On the branch considered here, reversing the angular velocity reverses frame dragging and leaves the shape unchanged. Dragging therefore begins at first order, while deformation begins at second order. We treat the dragging here and the deformation in Section 5.

An isolated shell whose particles follow its globally timelike stationary symmetry is static under the hypotheses of Appendix C, even if its shape is not spherical. Internal motion is therefore necessary for the stationary dragging studied here. The appendix also restricts stationary label motion near relaxation to Euclidean rotations,

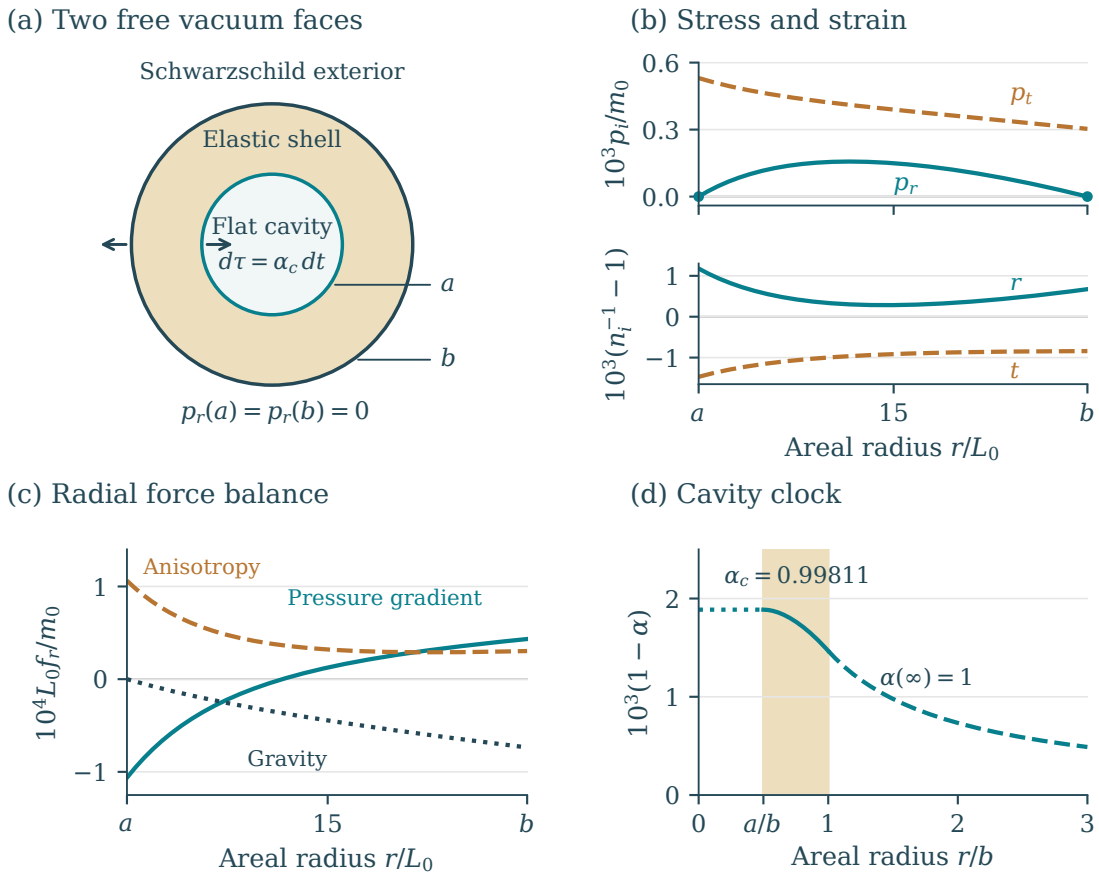

Figure 1. Static support for the elastic shell of Eq. (3.5). (a) Areal-coordinate section with outward material normals at both free faces. The flat cavity has lapse α_c relative to infinity. (b) Radial and tangential pressures (positive in compression) and departures of the proper stretches $1/n_r = d\ell/dR$ and $1/n_t = r/R$ from unity. Free radial traction permits nonzero tangential stress. (c) Pressure-gradient, gravity and anisotropy terms of Eq. (2.4), per unit areal radius. Outward is positive; their sum vanishes. Anisotropy supplies outward support near a. (d) Proper clock rate $\alpha = d\tau/dt$, with t normalized at infinity. Shading marks the material.

but that kinematic result does not establish rigid rotation in physical space. For rigid rotation we use the following existence result.

In rigid rotation all particles have the same angular velocity, and their strain is constant along their worldlines. The equilibrium shape can still depend on that angular velocity.

Theorem 4.1 (Slowly rotating elastic shells [8]). *Fix the law (2.6) with positive moduli and $\kappa+4\mu_0/3 < m_0$, and a round relaxed material annulus $A < |X| < B$. For sufficiently small $G > 0$ and $|\varpi|$, there is a stationary, axisymmetric Einstein–elastic shell near relaxation with regular vacuum cavity and asymptotically flat exterior. Both faces have zero traction, and*

$$u \parallel K + \varpi L, \qquad [K, L] = 0, \tag{4.1}$$

where K is normalized at infinity and L has 2π-periodic axial orbits. The matter obeys strict DEC and has causal material characteristics. The solution is locally unique within the rigid-rotation construction after fixing its gauge and rigid motions.

The annular-domain and constitutive hypotheses, including smooth parameter dependence in material coordinates, are given in Appendix D.

The cavity remains regular vacuum but need not remain flat; the rotating exterior is generally not Schwarzschild. The existence neighborhood is unquantified and need not contain the numerical shell at its stated parameters. The first-order response about that sphere can nevertheless be calculated without assuming a finite rotating continuation.

The linear stationary rotational equation about a spherical equilibrium is more explicit. With ϖ the angular velocity and $\omega(r) = O(\varpi)$,

$$ds^2 = -e^{2\Phi}dt^2 + e^{2\Lambda}dr^2 + r^2 d\Omega^2 - 2r^2 \sin^2\theta\, \omega(r)\, dt\, d\phi + O(\varpi^2). \tag{4.2}$$

The axial momentum equation is

$$(r^4 j\omega')' = -16\pi G r^4 e^{\Lambda-\Phi}(\rho + p_t)(\varpi - \omega), \qquad j = e^{-\Phi-\Lambda}. \tag{4.3}$$

To obtain this equation, the shift in the convention of Appendix H is $\beta^\phi = -\omega$. Its only independent nonzero extrinsic-curvature component is

$$K_{r\phi} = -\tfrac{1}{2} r^2 \sin^2\theta\, e^{-\Phi}\omega'.$$

Its trace vanishes. The material momentum density is

$$S_\phi = (\rho + p_t) r^2 \sin^2\theta\, e^{-\Phi}(\varpi - \omega).$$

Substituting these expressions into $D_j K^j{}_\phi = 8\pi G S_\phi$ gives Eq. (4.3), including its sign and normalization. The source contains the tangential enthalpy $\rho + p_t$: the material's energy density and tangential pressure both contribute to rotational inertia. Regularity gives $\omega' = 0$ in the cavity; asymptotic nonrotation gives $\omega = 2J/r^3$ in the exterior at this order, with J the geometrized ADM angular momentum.

Let $f(a) = 1$, $f'(a) = 0$ solve

$$(r^4 j f')' = 16\pi G r^4 e^{\Lambda-\Phi}(\rho + p_t) f, \qquad C = f(b) + \tfrac{b}{3} f'(b). \tag{4.4}$$

Then $\varpi - \omega = \varpi f/C$ in the material and

$$\frac{\omega_{\rm cav}}{\varpi} = 1 - \frac{1}{C}, \qquad \frac{J}{\varpi} = \frac{b^4 f'(b)}{6C} > 0. \tag{4.5}$$

Indeed, the positive right-hand side gives $f' > 0$ beyond the inner face and $C > f(b) > 1$. Hence $0 < \omega < \varpi$ for $\varpi > 0$: the cavity's inertial frame turns in the same sense as the material, at a smaller angular rate. Within the cavity, $T = \alpha_c t$ and $\widetilde{\phi} = \phi - \omega_{\rm cav} t$ make the metric Minkowskian through first order. Its relative rotation with respect to infinity is fixed by matching through the shell. Local flatness at this order therefore coexists with a measurable dragging response; curvature enters at the next order. For the numerical spherical example,

$$\omega_{\rm cav}/\varpi \simeq 0.002515, \qquad J/\varpi \simeq 5.186\, L_0^3. \tag{4.6}$$

Figure 2(a) shows this linear response.

The weak-gravity limit connects dragging directly to the static redshift. At fixed (β, k, μ) the relaxed density is m_0, the equilibrium pressures are $O(\chi m_0)$, and $a = A + O(\chi A)$, $b = B + O(\chi A)$. To leading order, the cavity potential and the dragging equation reduce to

$$\Phi_c = -4\pi G m_0 \int_A^B R\, dR + O(\chi^2), \qquad f'(r) = \frac{16\pi G m_0}{5r^4}(r^5 - A^5) + O(\chi^2/A). \tag{4.7}$$

The first expression is the Newtonian potential at any interior point of a spherical annulus. The second follows by integrating $(r^4 f')' = 16\pi G m_0 r^4$ with $f'(A) = 0$. Substitution into $C = f(B) + B f'(B)/3$ gives

$$\begin{aligned} 1 - \alpha_c &= 2\pi\chi(\beta^2 - 1) + O(\chi^2), \\ \frac{\omega_{\rm cav}}{\varpi} &= \frac{8\pi}{3}\chi(\beta^2 - 1) + O(\chi^2) = \frac{4}{3}(1 - \alpha_c) + O(\chi^2). \end{aligned} \tag{4.8}$$

Here $\omega_{\rm cav}/\varpi$ denotes the linear rotational response; both angular rates use time normalized at infinity. For a relaxed mass layer $d\mathcal{M}_0 = 4\pi m_0 R^2 dR$, the common radial weight is $G\, d\mathcal{M}_0/R$. Thus both observables measure an inverse-radius mass moment. This is the weak-gravity form of the shell-induced dragging discussed in Ref. [17]. The factor 4/3 comes from the rotational field equation and vacuum matching; it is independent of the selected elastic moduli at this order. It is not an exact relation at finite compactness. For $\chi = 10^{-4}$ and $\beta = 2$, the leading redshift and dragging fraction are 0.001885 and 0.002513, respectively.

The dragging solution also determines the moment of inertia and the leading rotational energy. Let $\mathcal{M} = M/G$ and $\mathcal{J} = J/G$ be the physical mass and angular momentum. Along the smooth branch at fixed material content, $d\mathcal{M} = \varpi\, d\mathcal{J}$ (Appendix F). Reversal of rotation makes $\mathcal{M}$ even and $\mathcal{J}$ odd. Hence

$$\begin{aligned} \mathcal{J} &= \mathcal{I}_G \varpi + O(\varpi^3), \qquad \mathcal{M} = \mathcal{M}_G + \tfrac{1}{2}\mathcal{I}_G \varpi^2 + O(\varpi^4), \\ \mathcal{I}_G &= \frac{8\pi}{3}\int_a^b r^4 e^{\Lambda - \Phi}(\rho + p_t)\frac{f}{C}\, dr > 0. \end{aligned} \tag{4.9}$$

Here $\mathcal{I}_G$ is the moment of inertia measured from the asymptotic angular momentum. The last identity follows by integrating Eq. (4.4). At $G = 0$, $\mathcal{I}_0 = 8\pi m_0(B^5 - A^5)/15$. Thus the dragging response also fixes the leading rotational energy on the existence branch. The first-order calculation at the numerical example does not, by itself, establish such a branch there.

4.1 A circular clock comparison

A difference in arrival times gives an operational measure of this rotation. Send two light signals in opposite directions around a fixed loop, and compare their returns on the emitter's own clock. To describe the experiment, use coordinates adapted to the guide. Where their stationary Killing field is timelike, write

$$g = -e^{2U}(dt + \psi_i dx^i)^2 + e^{-2U}h_{ij}dx^i dx^j, \qquad e^{2U} > 0, \quad h > 0. \tag{4.10}$$

For two light signals sent in opposite directions around the same fixed spatial loop $\mathcal{C}$, the future-directed null condition gives

$$dt = -\psi_i dx^i + e^{-2U}\sqrt{h_{ij}dx^i dx^j}, \qquad \Delta\tau = -2e^{U(x_0)}\oint_{\mathcal{C}} \psi_i dx^i, \tag{4.11}$$

where x_0 is the emitter and stationary mirrors or guides maintain the loop. The symmetric path-length term cancels between the two circuits. The delay is invariant under $t \mapsto t + f(x)$ and vanishes for a guide at rest in the static shell, where $\psi = 0$.

For an equatorial circular guide at radius r, let Ω_g be its angular velocity relative to infinity. In coordinates rotating with the guide, $g_{t\phi}$ becomes $g_{t\phi} + \Omega_g g_{\phi\phi}$ and g_{tt} becomes $g_{tt} + 2\Omega_g g_{t\phi} + \Omega_g^2 g_{\phi\phi}$. Equation (4.11) therefore gives

$$\Delta\tau = \frac{4\pi(g_{t\phi} + \Omega_g g_{\phi\phi})}{\sqrt{-g_{tt} - 2\Omega_g g_{t\phi} - \Omega_g^2 g_{\phi\phi}}}, \tag{4.12}$$

provided $g_{\phi\phi} > 0$ and the guide worldlines are timelike. The sign compares the positive azimuthal circuit with the negative one. To first order in ϖ and Ω_g, this is $4\pi r^2 e^{-\Phi}(\Omega_g - \omega)$. A guide corotating with the material has $\Omega_g = \varpi$ and includes the kinematic Sagnac delay. A guide fixed relative to infinity has $\Omega_g = 0$ and measures the term proportional to $-\omega$. In the cavity these are ideal test guides whose stress-energy is neglected. A guide following the local inertial axes, $\Omega_g = \omega_{\rm cav}$, has zero delay at this order.

5 Deformation, cavity tides, and the central clock

At first order, rotation drags the cavity's inertial frame without producing tidal curvature there. At second order, deformation and stress determine the leading tide and change the central clock rate. We calculate both from the same material response.

5.1 Centrifugal deformation

The leading deformation in weak gravity follows from the elastic problem at $G = 0$. This is a special-relativistic calculation: distances between neighboring material particles are measured in their local rest frames. The resulting correction enters at order ϖ^2, the same order as the displacement [18]. It must be retained even though the angular velocity is small. On the reference annulus $\mathcal{D} = \{X : A < |X| < B\}$, let $x = R_z(\varpi t)\phi(X)$, where R_z is a rotation about the z axis. For this calculation, put $F = D\phi$, $v = \varpi e_z \times \phi$, and $X_\perp = (X^1, X^2, 0)$. The rest-space strain and energy per reference volume are

$$C = F^{\mathsf{T}}\left(I + \frac{vv^{\mathsf{T}}}{1-|v|^2}\right)F, \qquad \epsilon(C) = \sqrt{\det C}\,\rho(n_i), \qquad n_i = \lambda_i(C)^{-1/2}. \tag{5.1}$$

The extra term is the Lorentz correction along the local direction of motion: the matrix in parentheses is the inverse of $I - vv^{\mathsf{T}}$, the spatial part of the inverse material strain. The stationary material action per unit time is $-\int_{\mathcal{D}} \sqrt{1-|v|^2}\epsilon(C)\,dX$.

Proposition 5.1 (Leading centrifugal response)**.** *At $G = 0$, write the centered, axially aligned branch as $\phi(X) = X + \varpi^2 u(X) + O(\varpi^4)$. Put $\lambda = \kappa - 2\mu_0/3$, $w = e_z \times X$ and $\tau(E) = \lambda \operatorname{tr}(E) I + 2\mu_0 E$. The coefficient u, modulo rigid motions, is the unique solution of*

$$\begin{aligned} \mu_0 \Delta u + (\lambda+\mu_0)\nabla \operatorname{div} u &= -(m_0 + \lambda - \mu_0)X_\perp, \\ \tau(e(u))n = -\tfrac{\lambda}{2}|X_\perp|^2 n \quad (|X| = A, B), &\qquad e(u) = \tfrac{1}{2}(Du + Du^{\mathsf{T}}). \end{aligned} \tag{5.2}$$

Here τ is positive in tension; the physical principal pressures have the opposite sign. The source excites only the spherical monopole and spheroidal quadrupole.

Proof. Equation (5.1) gives $C = I + 2\varpi^2 E + O(\varpi^4)$, where $E = e(u) + w \otimes w/2$. Since $D\epsilon(I) = 0$, the u-dependent coefficient of ϖ^4 in the reduced energy is

$$\int_{\mathcal{D}} \left[\frac{\lambda}{2}(\operatorname{tr} E)^2 + \mu_0 |E|^2 - m_0 X_\perp \cdot u\right] dX. \tag{5.3}$$

Variation gives $\operatorname{div}\tau(E) = -m_0 X_\perp$ and $\tau(E)n = 0$. Direct differentiation yields $\operatorname{div}\tau(w \otimes w/2) = (\lambda - \mu_0)X_\perp$; on either sphere, $w \cdot n = 0$. These identities give the displayed equations. The loads have zero total force and torque by axial and reflection symmetry. The positive elastic form and Korn's inequality [19] give existence and uniqueness on the rigid-motion quotient. The angular decomposition of $X_\perp$ and $|X_\perp|^2 n$ contains only the stated two harmonics. □

Write $s = |X|/A$ and $P_2 = (3\cos^2\theta - 1)/2$. The displacement has the form

$$u/A^3 = (U_0 + U_2 P_2)e_r + V_2 \partial_\theta P_2\, e_\theta. \tag{5.4}$$

The monopole U_0 changes the mean radius; the quadrupole U_2 changes the polar radius relative to the equatorial radius, and V_2 describes tangential material motion. Appendix E reduces the boundary problem to six linear equations and gives their exact solution for the selected moduli and $B = 2A$. A useful measure of each face's deformation is

$$\frac{r_{\mathrm{eq}} - r_{\mathrm{pole}}}{As} = \varepsilon_\Omega\, \mathcal{F}(s) + O(\varepsilon_\Omega^2), \qquad \mathcal{F}(s) = -\frac{3U_2(s)}{2s},$$
$$\mathcal{F}(1) \simeq 17.56, \quad \mathcal{F}(2) \simeq 8.54. \tag{5.5}$$

Both faces are oblate, with the larger fractional flattening at the inner face. The surfaces are not scaled copies of one another. These response coefficients depend on thickness and elastic moduli; they are not universal numbers. Their strict signs persist in a neighborhood of the selected parameters by continuous dependence of the invertible boundary problem. At small positive G, the shape increment has corrections $O(G\varpi^2 + \varpi^4)$ in fixed units.

5.2 The tide measured in the cavity

At first order in G, write $g_{00} = -(1 + 2\Phi_{\mathrm{lin}})$. The leading quadrupolar part of the potential in the regular cavity is $\Phi_2 = qr^2 P_2(\cos\theta)$. Its measurable content is the electric tidal tensor, which determines relative free-fall acceleration. In a locally inertial orthonormal frame at the center, $R_{\hat{0}\hat{i}\hat{0}\hat{j}} = \partial_i\partial_j\Phi_2 = q\,\mathrm{diag}(-1,-1,2)$, up to $O(G^2\varpi^2 + G\varpi^4)$ on compact subsets of the cavity. Let $\xi^{\hat{i}}$ be an initially comoving tracer's separation from the central geodesic and τ the latter's proper time. Geodesic deviation gives, to this order,

$$\frac{d^2}{d\tau^2}\begin{pmatrix}\xi^{\hat{x}}\\ \xi^{\hat{y}}\\ \xi^{\hat{z}}\end{pmatrix} = q\begin{pmatrix}\xi^{\hat{x}}\\ \xi^{\hat{y}}\\ -2\xi^{\hat{z}}\end{pmatrix}. \tag{5.6}$$

Thus the sign of q distinguishes stretching along the rotation axis from stretching in the equatorial plane. The corresponding error in the dimensionless tensor $A^2 R_{\hat{0}\hat{i}\hat{0}\hat{j}}$ is $O(\chi^2\varepsilon_\Omega + \chi\varepsilon_\Omega^2)$ at fixed (β, k, μ). No numerical bound on its coefficient is assumed.

The stationary Einstein equation for this potential is $\Delta\Phi_{\mathrm{lin}} = 4\pi G(T_{00} + T_{ii})$. Indeed, stationarity gives $R_{00} = \Delta\Phi_{\mathrm{lin}}$ at linear order, and the trace-reversed Einstein equation gives $R_{00} = 4\pi G(T_{00} + T_{ii})$ at the same order. Here T_{ii} is summed over the three spatial directions: pressure as well as energy density sources this potential. Pulling the source back to the reference annulus gives

$$(T_{00} + T_{ii})(\phi)\det F = m_0 + \varpi^2\left[-3\kappa\,\mathrm{div}\,u + \tfrac{3}{2}(m_0 - \kappa)|X_\perp|^2\right] + O(\varpi^4). \tag{5.7}$$

This includes rest density, stress, kinetic energy and the moving volume. It follows from $\rho = m_0[1 - \varpi^2\,\mathrm{tr}\,E] + O(\varpi^4)$, $\sum p_i = -3\kappa\varpi^2\,\mathrm{tr}\,E + O(\varpi^4)$ and the two boost contributions $m_0\varpi^2|w|^2$ to T_{00} and T_{ii}.

To determine q from the material, expand the gravitational Green function. For $r < |x'|$, $|x - x'|^{-1} = \sum_{\ell \geq 0} r^\ell |x'|^{-\ell-1} P_\ell(\cos\gamma)$, where γ is the angle between x and x'. Axisymmetry therefore gives

$$q = -G \int_{\text{shell}} (T_{00} + T_{ii})(x') \frac{P_2(\cos\theta')}{|x'|^3} \, d^3x' \tag{5.8}$$

at first order in G. This is an inverse radial moment of the active source. At the same weak-field order, the exterior mass quadrupole weights the source by $|x'|^2$; the cavity is more sensitive to matter near the inner face. The spatially constant part of the cavity potential changes the central clock rate and drops out of the tidal tensor. We first isolate the quadrupole, then calculate that independent clock correction in Section 5.4.

Pull the integral back to the material annulus and expand its kernel as well as its source. Source transport contributes $m_0 u \cdot \nabla(R^{-3} P_2)$. The angular identities $\int P_2^2 d\Omega = 4\pi/5$ and $\int |\nabla_{S^2} P_2|^2 d\Omega = 24\pi/5$ give

$$\begin{gathered} \mathcal{T} = \int_1^\beta \frac{-3U_2 + 6V_2}{s^2} \, ds, \qquad \mathcal{C} = \int_1^\beta \frac{D_2}{s} \, ds, \qquad D_2 = U_2' + 2U_2/s - 6V_2/s, \\ q = -\frac{4\pi}{5} G m_0 A^2 \varpi^2 \left[\mathcal{T} - 3k\mathcal{C} - (1-k)\frac{\beta^2 - 1}{2} \right]. \end{gathered} \tag{5.9}$$

The first term accounts for the redistribution of relaxed mass by the displacement. The second comes from the pressure trace in Eq. (5.7). The last includes both motion and the pressure induced by the relativistic rest-space strain: $|X_\perp|^2 = 2R^2(1 - P_2)/3$ supplies its quadrupolar part. Here D_2 is the quadrupolar coefficient of the displacement divergence; positive D_2 denotes local volume expansion in that harmonic.

The definition of D_2 also gives $(-3U_2 + 6V_2)/s^2 = (U_2/s)' - D_2/s$. Integration and Eq. (5.5) now express the transport in terms of the faces:

$$\mathcal{T} = \left[\frac{U_2}{s} \right]_1^\beta - \mathcal{C} = \frac{2}{3} \big[\mathcal{F}(1) - \mathcal{F}(\beta) \big] - \mathcal{C}. \tag{5.10}$$

Combining the terms gives the quadrupole at order $G\varpi^2$ for any fixed annulus and positive causal moduli in the small-parameter branch:

$$\begin{aligned} q &= -G m_0 A^2 \varpi^2 \mathcal{Q}_c, \\ \mathcal{Q}_c &= \frac{4\pi}{5} \left\{ \frac{2}{3} \big[\mathcal{F}(1) - \mathcal{F}(\beta) \big] - (1 + 3k)\mathcal{C} - (1-k)\frac{\beta^2 - 1}{2} \right\}. \end{aligned} \tag{5.11}$$

The face term measures differential fractional flattening; the bulk term combines volume transport with its induced pressure. Shear stiffness enters through U_2, V_2. The response is fixed by the annulus and its elastic moduli. This decomposition uses the reference material coordinates of the weak-field construction. Its separate terms

describe contributions to one source integral; the observable is the resulting tidal tensor.

If $D_2 = 0$ and the faces have equal fractional flattening, the transport term vanishes even when both faces are oblate. This cancellation concerns transport only; the remaining active stresses still contribute to the tide. Oblateness alone does not determine its sign.

5.3 Two materials, opposite tides

At $(\beta, k, \mu) = (2, 1/5, 1/10)$, the three terms in braces in Eq. (5.11) are approximately 6.011, -1.322, and -1.200. Differential flattening dominates, so $\mathcal{Q}_c \simeq 8.77 > 0$ and $q < 0$ at leading order. Appendix E evaluates the integrals exactly. The coefficient depends on thickness and moduli; its strict sign persists locally by continuity of the invertible elastic boundary problem.

The sign can reverse within the same family of elastic laws while maintaining causality near relaxation. Keep $\beta = 2$ and choose $k = \mu = 2/5$. The relaxed squared sound speeds are $2/5$ and $14/15$, so the small-coupling existence argument still applies. The two faces remain oblate, with $\mathcal{F}(1) \simeq 3.06$ and $\mathcal{F}(2) \simeq 1.45$, but now $\mathcal{Q}_c \simeq -0.866$ and $q > 0$. The exact coefficients are given in Appendix E. Differential flattening is too small to overcome the bulk and motion terms. A linear interpolation between these two choices stays within the causal relaxed-modulus domain and therefore passes through a zero of the leading quadrupole coefficient. That cancellation leaves higher-order curvature undetermined; it does not establish an exactly flat rotating cavity.

For either explicit material, smooth dependence and the nonzero leading tidal coefficient rule out a flat cavity in a sufficiently small two-parameter neighborhood with $G > 0$ and $\varpi \neq 0$. The uniform remainder follows by factoring the central electric curvature by $G\varpi^2$: it vanishes at $G = 0$ and at $\varpi = 0$, is even in ϖ, and depends smoothly on both parameters in the regular cavity. The remaining coefficient is continuous and has the nonzero limit computed above. This establishes a neighborhood, rather than a numerical range of angular velocities. The first material stretches tracer separations along the rotation axis and focuses them in the equatorial plane; the stiffer material reverses these directions. This curvature cannot be removed by a change of coordinates.

The zero trace of the tidal tensor expresses $R_{\hat{0}\hat{0}} = 0$ in vacuum. For either sign of nonzero q, at least one direction stretches rather than restores the separation. The leading tide therefore supplies no three-dimensional passive confinement near the center. This is a statement about freely falling test particles; it does not decide whether the shell's elastic modes grow. At $q = 0$, higher orders must be examined.

Figure 2(b–d) shows both oblate faces, these relative accelerations, and the source contributions. Its exaggerated display amplitude illustrates the leading coefficients; it supplies no finite-rotation bound on strain or curvature.

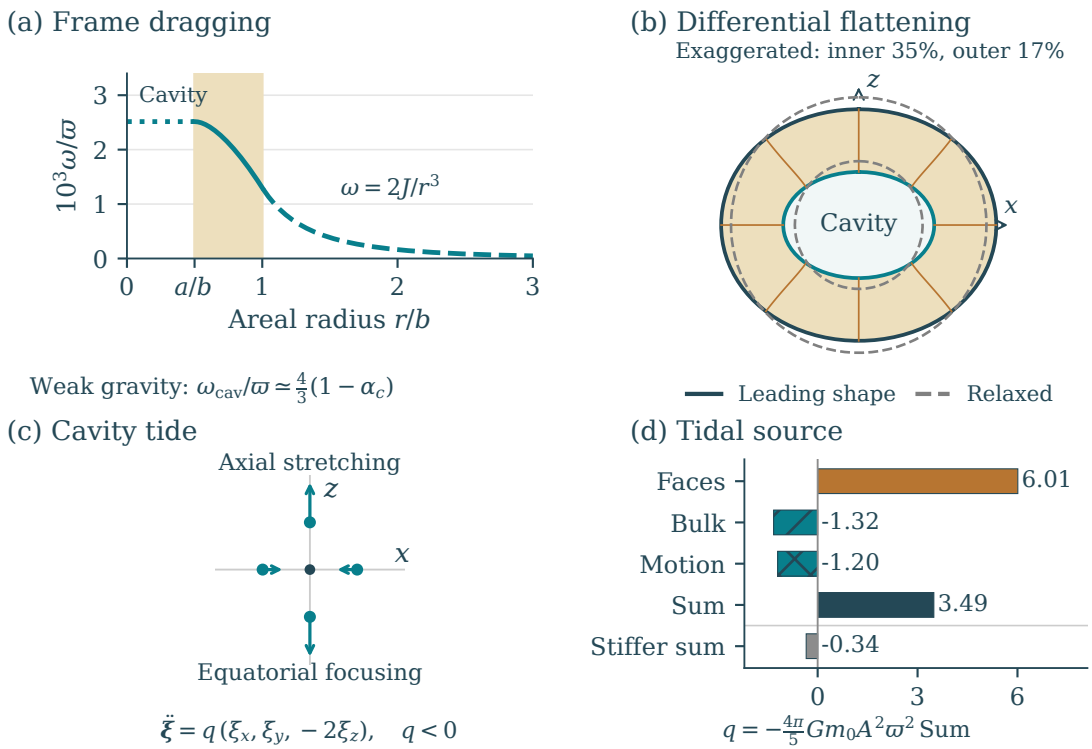

Figure 2. Rotational response of the shell in Eq. (3.5). (a) Frame dragging ω at first order in the material angular velocity ϖ, matched across both vacuum faces and measured relative to infinity. (b) Leading centrifugal displacement at $G = 0$, $B = 2A$, drawn at equal scales with exaggerated amplitude $\varepsilon_\Omega = (A\varpi)^2 = 0.02$. The fractional flattenings $\varepsilon_\Omega \mathcal{F}$ are 35% (inner) and 17% (outer), relative to the relaxed radii. Solid faces include radial and tangential displacement; dashed circles are relaxed faces, and interior curves follow material meridians. (c) Local relative free-fall acceleration at order $G\varpi^2$. Dots mark equal initial separations from the central geodesic; arrows share an arbitrary scale. (d) Dimensionless source terms in Eq. (5.11): Faces denotes differential fractional flattening; Bulk includes volume transport and pressure; Motion includes kinetic energy and the relativistic rest-space correction. Only the sum determines q. The stiffer material, $k = \mu = 2/5$, gives $q > 0$ at the same thickness despite two oblate faces. Panels (b–d) show leading coefficients without a finite-rotation error bound.

5.4 The rotation-induced central clock shift

Let $\alpha_{\rm cen}(G, \varpi) = \sqrt{-g(K, K)}$ at the center of the reflection-symmetric rotating shell, with K normalized at infinity. The center follows a geodesic by symmetry. Its proper clock rate is $d\tau/dt = \alpha_{\rm cen}$, and $\alpha_{\rm cen}(G, 0) = \alpha_c$. At leading order in gravity the Green function at the center gives

$$\Phi_{\rm lin}(0) = -G \int_{\rm shell} \frac{T_{00} + T_{ii}}{|x'|}\, d^3x'. \tag{5.12}$$

Pull back to material coordinates, as for the tide. The kernel changes by $\delta(1/|\phi|) = -\varpi^2 u_r/R^2$. Only the spherical averages contribute: $\langle u_r\rangle = A^3U_0$, $\langle \operatorname{div} u\rangle = A^2D_0$, where $D_0 = U_0' + 2U_0/s$, and $\langle \sin^2\theta\rangle = 2/3$. Using Eq. (5.7), we obtain

$$\alpha_{\rm cen}(G,\varpi) - \alpha_c = \chi\varepsilon_\Omega \mathcal{Z}_c + O(\chi^2\varepsilon_\Omega + \chi\varepsilon_\Omega^2),$$
$$\mathcal{Z}_c = 4\pi \int_1^\beta \left[U_0 + 3ksD_0 - (1-k)s^3\right] ds. \tag{5.13}$$

The terms arise from displacement of the mass, the pressure trace, and motion including the relativistic rest-space strain. Here a positive $\mathcal{Z}_c$ means that rotation makes the central clock run faster relative to infinity than in the static shell with the same material content. The asymptotic error is at fixed (β, k, μ).

The original material has $\mathcal{Z}_c = 16472\pi/2625 \simeq 19.71$. The stiffer choice $k = \mu = 2/5$ instead gives $\mathcal{Z}_c = -15929\pi/7350 \simeq -6.81$. Appendix E gives the elementary integral evaluation. Both materials gain rotational mass, yet their clock shifts have opposite signs. There is no conflict: the mass uses the unweighted active source, the central potential weights it by $1/|x'|$, and the tide weights its quadrupole by $P_2/|x'|^3$. Moving material outward can weaken the central potential even while the total energy increases. These leading coefficients do not give a finite-rotation error bound or a universal relation between the signs of the clock shift and the tide.

6 Dynamics and physical scope

An equilibrium and its response to steady rotation do not determine how the shell evolves after a disturbance. We now distinguish linear mode stability from smooth nonlinear evolution, then consider what uniform translation changes in the cavity measurements.

6.1 Linear perturbations

Radial perturbations expand or contract spherical layers. Odd-parity perturbations twist them, while nonspherical even-parity perturbations change their shape and density. These sectors decouple on the static spherical background, so stability in one does not decide the others. Stability requires an energy estimate for the coupled matter and gravitational field, including both moving vacuum faces. The interface conditions are

$$\Delta(T_{\mu\nu}n^\nu) = 0, \qquad [\Delta h_{ab}] = 0, \qquad [\Delta \mathcal{K}_{ab}] = 0, \tag{6.1}$$

where h_{ab} and $\mathcal{K}_{ab}$ are the induced metric and extrinsic curvature, brackets use the same normal on both sides, and Δ includes the surface displacement and normal variation. In spherical motion the material condition is $\Delta p_r = 0$, rather than a condition at a fixed coordinate radius.

For radial perturbations, eliminating the metric constraints gives a self-adjoint equation for the material displacement, with gravitational backreaction included. A positive energy bound then gives a lower bound on squared oscillation frequencies. The sufficient criterion in Eq. (I.5) holds at sufficiently small coupling; its sampled value for Eq. (3.5) is about $1.8\times10^{-3}L_0^{-2}$. For odd parity, positive tangential enthalpy and shear stiffness give a nonnegative coupled energy for every angular harmonic, including exterior gravitational waves. This excludes exponentially growing physical modes in that sector, but gives no positive lower bound on radiative frequencies or decay rate. These results are derived in Appendices I and J.

Nonspherical even-parity and rotating stability remain unresolved. For perfect-fluid thin shells with a flat interior, Ref. [20] finds growing even-parity shape modes at every sampled compactness, adiabatic index, and multipole $\ell \geq 2$. Elastic membranes on a prescribed Schwarzschild background can also be radially stable but unstable to dipolar displacements [21]. Neither result decides the stability of this finite-thickness, self-gravitating solid. Resolving its even-parity sector requires the coupled metric and elastic perturbations, with both faces displaced and matched to the two vacuum regions. The limit of vanishing thickness at fixed surface mass is singular: it changes the density scale, elastic response and interface equations. Neither these thin-shell instabilities nor the present stability estimates can be transferred across that limit without a separate derivation.

6.2 Smooth evolution and wave breaking

Conservation of ADM mass bounds a nonlinear radial energy norm while strain and velocity remain near equilibrium. Control over a prescribed finite time also requires small spatial derivatives and initial data compatible with the gravitational constraints and free faces. Under these conditions, the higher-regularity estimate controls the solution through reflections. The permitted size of the initial data depends on the chosen time interval.

Small amplitudes alone do not ensure this control. A sufficiently narrow pulse can have small strain and velocity but large gradients. Theorem I.5 constructs such radial pulses whose first derivatives diverge before either face is reached. Their steepening dominates the bounded gravitational sources while the state remains causal and obeys strict DEC. This is consistent with the finite-time estimate, whose H^s norm controls several spatial derivatives. The theorem concerns the first loss of smoothness, not an exponentially growing linear mode. Discontinuous continuation and long-time smooth evolution remain open.

6.3 Uniform motion and additional loads

Uniform translation gives an exact comparison beyond these perturbative calculations. Apply an asymptotic Lorentz boost to both the metric and material map.

This gives a coasting shell with $V^i = P^i/P^0$ and the same comoving cavity measurements. Appendix G gives the construction and its asymptotic hypotheses. The boost constructs uniform motion, not an acceleration process. Changing the shell's momentum requires exchange with another system or with outgoing radiation; the total ADM momentum of the complete isolated system remains conserved. Radiative steering [22] introduces photon emission and a different material law.

A massive payload changes the vacuum-cavity problem; supports holding it relative to the shell also introduce tractions. An external tidal field changes the gravitational boundary data even when the cavity stays empty. The response must be found from the coupled metric and elastic equations, as in relativistic tidal calculations for elastic crusts [23]. Love numbers characterize the induced exterior multipoles in response to an applied tidal field; the rotation-induced cavity tide calculated here answers a different question. The locally flat cavity of the isolated static solution does not establish gravitational shielding.

7 Conclusions

The leading cavity tide and rotational clock shift measure distinct source moments of an elastic shell. The tide depends on differential flattening of the two faces, deformation within the shell, and the stress and energy of rotation. The clock shift depends on an inverse-radius moment of the active source. For two choices of elastic moduli, each causal near relaxation, both observables reverse sign while both faces remain oblate and the total mass increases. The cavity geometry therefore resolves differences in material response that qualitative shape and total mass do not determine.

These results follow from the same constitutive action that supplies the shell's static support: tangential compression balances gravity while radial traction vanishes at both free faces. The static cavity is locally flat, with clocks redshifted relative to infinity. In weak gravity, this redshift and linear frame dragging measure the same inverse-radius mass moment: the dragging fraction $\omega_{\mathrm{cav}}/\varpi$ is $4/3$ times the clock-rate deficit $1 - \alpha_c$ at leading order. Tidal curvature first appears at second order in rotation. The leading vacuum tide has stretching and focusing directions and provides no three-dimensional passive confinement near the center.

The equilibrium and response calculations hold near relaxation. A complete linear stability analysis still requires the nonspherical even-parity sector with both moving faces and gravitational backreaction. Predictions at a specified angular velocity require a bound on the rotational remainder and on the resulting strain; the leading clock and tidal coefficients do not supply those bounds. A payload or an external tide changes the boundary-value problem. Energy conservation does not prevent wave breaking; evolution beyond that point requires an admissible weak-solution

theory or additional material physics. The construction describes the support, geometry, and partial stability of an ideal gravitating solid. A material realization and a process that forms or accelerates the shell require additional physics.

The sign change raises a concrete question: can the leading cavity tide vanish on a rotating branch stable to nonspherical perturbations? At a zero of $\mathcal{Q}_c$, the higher-order cavity curvature must also be determined. Together with bounds on strain and the rotational remainder, this would establish how nearly flat the cavity can remain over a finite range of angular velocities. A complementary inverse problem asks which combinations of bulk and shear response can be inferred from joint clock and tidal measurements for a known reference annulus and density. The observables probe distinct source moments, but uniqueness of this inference and its sensitivity to the assumed constitutive law remain to be determined.

A An exact smooth anisotropic shell

Theorem A.1 (Tangentially supported shell)**.** *Let $\rho \geq 0$ be a C^2 function supported in $[a, b]$, with $0 < a < b$, and define $m(r) = 4\pi \int_0^r \rho(s)s^2 ds$, $M = m(b)$. Suppose $2m(r) < r$ for $r > 0$. Set*

$$p_r = 0, \qquad p_t = \frac{\rho m}{2(r - 2m)}, \qquad \Phi(r) = -\int_r^\infty \frac{m(s)}{s[s - 2m(s)]}\, ds. \tag{A.1}$$

Then Eq. (2.1) solves all Einstein equations. The cavity is flat and redshifted, the exterior is Schwarzschild, and neither interface has a surface layer. Where $\rho > 0$, DEC holds if and only if $5m \leq 2r$. NEC and WEC hold everywhere.

Proof. The first two equations in Eq. (2.2) follow directly. With $p_r = 0$, Eq. (2.3) becomes $2p_t/r = \rho\Phi'$, which is exactly Eq. (A.1). Thus the angular equation holds as well. In the cavity $m = 0$ and Φ is constant. Outside b, $m = M$ and $e^{2\Phi} = 1 - 2M/r$. The prescribed regularity and $p_r = 0$ give smooth matching. Both ρ and p_t are nonnegative, and $p_t \leq \rho$ reduces to $5m \leq 2r$ where $\rho > 0$. At zero density all stresses vanish. □

A compact example is obtained directly from the mass profile

$$m(r) = MH\left(\frac{r - a}{b - a}\right), \qquad H(s) = 35s^4 - 84s^5 + 70s^6 - 20s^7 \quad (0 < s < 1), \tag{A.2}$$

extended by $H = 0$ to the left and $H = 1$ to the right. Since $H' = 140s^3(1 - s)^3$, the density is nonnegative and C^2 at both interfaces. The sufficient global bound $5M < 2a$ ensures DEC. A C^∞ example follows by replacing H' by a normalized positive bump $\exp[-1/(s(1 - s))]$ on $(0, 1)$; the same compactness bound applies. For example, $(a, b, M) = (10, 20, 1)$ satisfies this bound. This exact construction establishes that a vacuum interface need not violate DEC.

An Einstein-cluster interpretation is available when the supporting particles have timelike circular orbits. Their local squared speed is

$$v_{\mathrm{circ}}^2 = r\Phi' = \frac{m}{r-2m} < 1 \quad \Longleftrightarrow \quad 3m < r. \tag{A.3}$$

An isotropic distribution of orbital planes has $p_t = \rho v_{\mathrm{circ}}^2/2$, reproducing Eq. (A.1). This ideal distribution is concentrated on circular orbits; it is not a smooth distribution on the full particle phase space. The stability results in Section 6 concern the elastic action (2.6), which specifies the response to perturbations.

A.1 An inner pressure jump

The obstruction in Proposition 2.1 cannot be removed by truncating a positive-pressure fluid solution. Integrating Eq. (2.5) inward from $p(b) = 0$ generally produces $p(a) > 0$. Replacing that value by zero in the cavity creates a new matching problem. At a sharp inner interface, $m(a) = 0$ gives $\Phi'(a^+) = 4\pi a p(a^+)$ while the vacuum cavity has $\Phi'(a^-) = 0$. With the normal pointing toward increasing r, the Israel surface tensor has $\sigma = 0$ and tangential pressure $P_\Sigma = ap(a^+)/2$. Thus a nonzero positive inner pressure produces a massless surface layer that violates surface DEC. Smoothing the jump instead produces a transition stress that must be included explicitly; it does not retain the original isotropic equilibrium.

B Explicit constitutive bounds

The local sound speeds follow from two quadratic forms: the kinetic matrix W gives the inertia of a material disturbance, and $H(k)$ gives its restoring stiffness for wave covector k. Their generalized eigenvalues are the squared sound speeds. We derive both matrices and bound them uniformly over the strain range used in the example.

Fix $\kappa/m_0 = 1/5$, $\mu_0/m_0 = 1/10$, and divide all densities, stresses, and acoustic matrices in this appendix by m_0. Write $(a, b, c) = (n_1, n_2, n_3)$ and $n = abc$. For a positive strain matrix B,

$$s^2 = \frac{\operatorname{tr}\mathsf{B}\operatorname{tr}(\mathsf{B}^{-1}) - 9}{12}, \qquad \widehat{\rho} = n + \frac{(n-1)^2}{10} + \frac{ns^2}{10}. \tag{B.1}$$

Here B has eigenvalues n_i^2; the deformation tensor C in Eq. (5.1) has reciprocal eigenvalues. The invariant is unchanged by inversion and is smooth also at repeated eigenvalues. Its principal stress, with cyclic permutations for the others, is

$$\widehat{p}_1 = \frac{a^3b/c + a^3c/b + 6a^2b^2c^2 - 6 - b^3c/a - bc^3/a}{60}. \tag{B.2}$$

For the spherical equations, the same law reduces to

$$\begin{aligned} \widehat{\rho}(\nu,\eta) &= \frac{\nu^4 + 6\nu^3\eta^4 + 46\nu^2\eta^2 + 6\nu + \eta^4}{60\nu}, \\ \widehat{p}_r(\nu,\eta) &= \frac{\nu^4 + 3\nu^3\eta^4 - 3\nu - \eta^4}{30\nu}, \qquad \widehat{p}_t(\nu,\eta) = -\frac{\nu^4 - 6\nu^3\eta^4 + 6\nu - \eta^4}{60\nu}. \end{aligned} \tag{B.3}$$

The spatial acoustic Hessian follows by substituting $F = \mathrm{diag}(a,b,c) + \epsilon\xi k^{\mathsf{T}}$ in Eq. (B.1), with $\mathsf{B} = FF^{\mathsf{T}}$, and taking $\partial_\epsilon^2 \widehat{\rho}|_{\epsilon=0} = \xi^{\mathsf{T}} H(k)\xi$. For the kinetic term use $\mathsf{B} = FF^{\mathsf{T}} - \dot{X}\dot{X}^{\mathsf{T}}$. Then W is diagonal, $W_{ii} = (\widehat{\rho} + \widehat{p}_i)/n_i^2$. Define

$$\begin{aligned}
L(a,b,c) &= \frac{3ab/c + 3ac/b + 12b^2c^2 + b^3c/a^3 + bc^3/a^3}{60},\\
T(a,b,c) &= \frac{ab/c + 2ac/b + 2bc/a + c^3/(ab)}{60},\\
B(a,b,c) &= \frac{a^2/c + b^2/c + 12abc^2 + 2c - a^2c/b^2 - b^2c/a^2}{60},\\
W_1(a,b,c) &= \frac{1}{120}\big[3ab/c + 3ac/b + 24b^2c^2 + b^3/(ac)\\
&\qquad + 90bc/a + c^3/(ab) - b^3c/a^3 - bc^3/a^3\big].
\end{aligned} \tag{B.4}$$

The entries are $H_{11} = L(a,b,c)k_1^2 + T(a,b,c)k_2^2 + T(a,c,b)k_3^2$, $H_{12} = B(a,b,c)k_1k_2$, and $W_{11} = W_1(a,b,c)$; simultaneous permutations of densities and indices give the remaining entries. In particular, $H_0 = |k|^2 I/10 + 7kk^{\mathsf{T}}/30$ and $W_0 = I$ at relaxation.

Here are rational bounds valid throughout $l \le a,b,c \le u$, where $l = 99/100$ and $u = 101/100$:

quantity	lower bound	upper bound
L	$19/60$	$7/20$
T	$19/200$	$21/200$
B	$11/50$	$37/150$
W_1	$24/25$	$26/25$
$\widehat{p}_1$	$-1/100$	$1/100$

(B.5)

The table follows by bounding each Laurent monomial separately. For a positive monomial $a^i b^j c^k$, put l in factors with positive exponents and u in those with negative exponents for its lower bound; reverse the choices for its upper bound. Reverse the bounds for a negative coefficient and add. For example,

$$\frac{19}{60} \le \frac{6l^2/u + 12l^4 + 2l^4/u^3}{60} \le L \le \frac{6u^2/l + 12u^4 + 2u^4/l^3}{60} \le \frac{7}{20}. \tag{B.6}$$

The other rows follow by the same substitutions in the displayed formulas. The resulting inequalities hold simultaneously for all states in the box.

For $|k| = 1$, the table bounds each diagonal entry of $H - H_0$ in magnitude by $1/60$ and each off-diagonal entry by $|k_ik_j|/75$. Since $\sum_{j\neq i} |k_ik_j| \le 1$, every absolute row sum is at most $1/60 + 1/75 = 3/100$. The symmetric matrix therefore satisfies

$$\frac{7}{100} I \preceq H \preceq \frac{109}{300} I, \qquad \frac{24}{25} I \preceq W \preceq \frac{26}{25} I. \tag{B.7}$$

The generalized Rayleigh quotient lies between $7/104 > 1/16$ and $109/288 < 2/5$, proving the characteristic bound in Eq. (3.6) for every polarization and direction. Also $\widehat{\rho} \ge n \ge (99/100)^3$ and $|\widehat{p}_i| \le 1/100$, so the DEC slack exceeds $24/25$.

C Stationary material motion and staticity

Stationarity means that the metric and stress do not change along a time symmetry. Individual particles can still move, as they do in a rotating body [8]. For an elastic solid, the material metric restricts this motion because it records which particles are neighbors.

Let $q_{ab} = g_{ab} + u_a u_b$ be the material rest metric and $b_{ab} = k_{AB}\partial_a X^A \partial_b X^B$ the material metric pulled back to spacetime. On the three-dimensional rest space, define $\mathsf{B}^a{}_b = q^{ac}b_{cb}$ and the spatial stress $\mathsf{S}^a{}_b = q^{ac}T_{cd}q^d{}_b$. The eigenvalues of B are n_i^2. Isotropy gives a smooth matrix function $\mathsf{S} = \mathcal{S}(\mathsf{B})$.

Proposition C.1 (Stationary motion of material labels)**.** *Consider an Einstein–elastic solution with the law (2.6) and $\kappa, \mu_0 > 0$. Let the particle worldlines of a single body be bijectively labeled by a bounded connected Euclidean domain $\mathcal{D}$ with smooth boundary. Suppose a complete spacetime Killing field K preserves the material worldtube, $\rho + p_i > 0$, and all strains lie in a sufficiently small fixed rotation-invariant neighborhood of $\mathsf{B} = I$. Then, in material coordinates centered at the centroid of $\mathcal{D}$,*

$$KX^A = \Omega^A{}_B X^B, \qquad \Omega_{AB} = -\Omega_{BA}, \tag{C.1}$$

where Ω is constant and its rotations preserve $\mathcal{D}$. If $\mathcal{D}$ has no continuous rotational symmetry, then $KX^A = 0$.

Proof. First, the stress determines the strain on this constitutive branch. Differentiating Eq. (2.6) at relaxation gives, for every symmetric matrix E,

$$D\mathcal{S}_I[E] = \mu_0 E + \left(\frac{\kappa}{2} - \frac{\mu_0}{3}\right)\operatorname{tr}(E) I. \tag{C.2}$$

For diagonal E, substitute $n_i = 1 + \epsilon E_{ii}/2$ to first order; orthogonal diagonalization and isotropy give the formula for general E. The eigenvalues of this linear map on traceless and isotropic strains are μ_0 and $3\kappa/2$. Thus $\langle E, D\mathcal{S}_I[E]\rangle \geq c|E|^2$ for some $c > 0$, using the matrix inner product $\langle E, F\rangle = \operatorname{tr}(E^{\mathsf{T}}F)$. Continuity preserves a positive lower bound in a sufficiently small ball about I. Integrating the derivative along each straight segment in that ball proves that $\mathcal{S}$ is injective there. The ball can be chosen rotation invariant and contained in the positive strain matrices.

The Einstein equations imply $\mathcal{L}_K T = 0$. Since $\rho + p_i > 0$, $-\rho$ is distinct from every spatial stress eigenvalue; hence the future unit timelike eigenvector u is unique and $\mathcal{L}_K u = 0$. The rest metric and spatial stress are therefore stationary. Isotropy and injectivity of $\mathcal{S}$ imply $\mathcal{L}_K \mathsf{B} = 0$, and consequently $\mathcal{L}_K b = 0$.

Because $[K, u] = 0$, the flow of K maps entire particle worldlines to particle worldlines. It descends through their labels to a flow on $\mathcal{D}$. The equality $\mathcal{L}_K(X^* k) = 0$ makes this a Euclidean isometry. Its generator ξ obeys $\partial_A \xi_B + \partial_B \xi_A = 0$. Differentiating and cycling the indices gives $\partial_A \partial_B \xi_C = 0$; connectedness then gives $\xi = c + \Omega X$

with constant antisymmetric Ω. These isometries preserve the bounded domain and its volume, so they fix its centroid. In centered coordinates $c = 0$, proving the claim. □

For a round material annulus, the allowed label motion is a rotation. In stationary coordinates $K = \partial_t$, Eq. (C.1) integrates to $X(t, x) = e^{t\Omega} f(x)$. Where f is invertible, particle paths are images under f^{-1} of material circles. Their spatial shapes and speeds need not be those of rigid rotation. The proposition therefore imposes no axial spacetime symmetry and supplies no additional circulating equilibrium.

The sector $KX = 0$ has a stronger geometric restriction. Here the particles follow the stationary time symmetry itself.

Proposition C.2 (Staticity of a comoving isolated shell)**.** *Suppose an isolated shell governed by the Einstein–elastic equations admits the decomposition (4.10) globally on $\mathbb{R} \times \mathbb{R}^3$, with time-independent fields, $K = \partial_t$, $e^{2U} > 0$, and positive-definite h. Assume a regular vacuum cavity, piecewise smooth fields with Darmois matching and no surface layers, and $U, h - \delta, \psi = O(r^{-1})$ with first derivatives $O(r^{-2})$ at infinity. If $KX^A = 0$ throughout the material, then $d\psi = 0$ and the spacetime is static.*

Proof. Put $F_{ij} = \partial_i \psi_j - \partial_j \psi_i$, with its indices raised using h. The inverse metric and volume element are

$$g^{-1} = -e^{-2U}\partial_t^2 + e^{2U} h^{ij} (\partial_i - \psi_i \partial_t)(\partial_j - \psi_j \partial_t), \qquad \sqrt{-g} = e^{-2U}\sqrt{h}. \tag{C.3}$$

When $\partial_t X = 0$, both the strain $H^{AB} = e^{2U} h^{ij} \partial_i X^A \partial_j X^B$ and the material volume term are independent of ψ. The curvature identity for Eq. (4.10) is [8]

$$R_g \sqrt{-g} = \sqrt{h} \left(R_h + 2\Delta_h U - 2|DU|_h^2 + \tfrac{1}{4} e^{4U} F_{ij} F^{ij} \right). \tag{C.4}$$

Varying ψ in the Einstein–material action therefore gives the source-free mixed equation

$$D_i (e^{4U} F^{ij}) = 0. \tag{C.5}$$

Multiply by ψ_j and integrate over the full spatial domain, including the cavity. Antisymmetry gives $(D_i \psi_j) F^{ij} = F_{ij} F^{ij}/2$, hence

$$\frac{1}{2} \int_{\mathbb{R}^3} e^{4U} F_{ij} F^{ij} \, dV_h = \lim_{R \to \infty} \int_{S_R} \psi_j e^{4U} F^{ij} n_i \, dA_h = 0. \tag{C.6}$$

The surface term is $O(R^{-1})$. At each material face the two fluxes cancel by the weak Einstein equation and the absence of a surface layer. The regular center contributes no boundary term. Positivity of h and e^{4U} implies $F = 0$. Since the spatial domain is $\mathbb{R}^3$, $\psi = df$ globally, and $T = t + f(x)$ removes the cross term in Eq. (4.10). □

This result allows arbitrary shell shape and anisotropic stress. It uses strict stationarity, a regular center, the global product topology, and asymptotically vanishing connection. It does not require stress injectivity. The rotating branch evades its comoving hypothesis: the particles follow $K + \varpi L$, not K. That helical generator becomes spacelike far from the axis, so it cannot replace K in the global positive metric decomposition above.

D The small rotating branch

Proof of Theorem 4.1. Theorem 4.6 of Ref. [8] applies to a bounded connected axisymmetric material domain with smooth boundary. It does not require that the domain contain the origin or that its boundary be connected. The annulus therefore qualifies. The law depends only on the principal densities, so it is isotropic and invariant under rotations of the material reference frame, as required in the rotating construction. The relaxed specific energy is $m_0 > 0$, and its elastic quadratic form is $\kappa(\operatorname{tr} e)^2 + 2\mu_0 |\operatorname{dev} e|^2 > 0$ for nonzero symmetric e. These are the required constitutive conditions. The construction's gauge anchor can be chosen on the rotation axis within the annulus, for example at $X_0 = (0, 0, (A+B)/2)$. The theorem solves the full equations and restores the projected force and gauge equations. Its small deformation preserves both boundary components and the vacuum cavity. DEC, the timelike material flow, and the characteristic gaps persist from relaxation by continuity on the compact body. The branch depends smoothly on G and ϖ in material coordinates. Indeed, the fixed-domain residual in Section 3 of Ref. [8] is smooth for this law: its coefficients use smooth strain invariants, matrix inverses and determinants near positive matrices. A bounded extension of the material displacement to all of $\mathbb{R}^3$, equal to zero outside a compact set, fixes both interfaces when the equations are pulled back. The unknowns lie in $W^{2,p}$ on the body and weighted $W^{2,p}_\delta$ spaces for the gravitational fields, with $p > 3$ and $-1 < \delta < -1/2$. Their first derivatives are bounded, so the products of first derivatives and coefficients multiplying second derivatives define smooth maps into the corresponding L^p spaces. The traction residual takes values in $W^{1-1/p,p}$ on both faces. The material indicator is fixed under parameter differentiation; one does not differentiate a moving density jump in a fixed spatial chart. The projected derivative is an isomorphism after removing rigid motions, so the smooth implicit-function theorem applies; the subsequent equilibration restores the full equations.

The chosen point fixes translations for the existence proof. For the cavity observables we translate along the axis to center the image of the material annulus, fixing $\int_{\mathcal{D}} \phi(X)\, dX = 0$, and fix its orientation. These conditions remove the same rigid-motion kernel. Equatorial reflection preserves the centered data, so local uniqueness makes it a symmetry of the branch. Together with axial symmetry it fixes the central worldline and sets its spatial acceleration to zero. Time reversal changes ϖ to $-\varpi$;

scalar clock rates and the central electric tidal eigenvalues are consequently even in ϖ. On compact subsets of the cavity, the stationary vacuum equations are elliptic in the chosen gauge. Interior regularity applied to their parameter derivatives promotes the smooth dependence above to the curvature and clock observables used in the main text. □

E Centrifugal response coefficients

The angular decomposition reduces the displacement problem to radial equations. The tractions on both faces fix their integration constants. An integrated force balance first relates the stresses to the rotational mass. Dot $\operatorname{div}\tau(E) = -m_0 X_\perp$ with X and integrate over the reference annulus. The free-traction terms vanish at both faces, giving

$$3\kappa \int_{\mathcal{D}} \operatorname{div} u \, dX = \left(m_0 - \frac{3\kappa}{2} \right) \int_{\mathcal{D}} |X_\perp|^2 \, dX. \tag{E.1}$$

Substitution in Eq. (5.7) gives the integrated order-ϖ^2 term

$$\tfrac{1}{2} m_0 \varpi^2 \int_{\mathcal{D}} |X_\perp|^2 dX = \tfrac{1}{2} \mathcal{I}_0 \varpi^2,$$

as required by the first law. The spatial stresses cancel in the unweighted integral of T_{ii} at this order. They need not cancel in the inverse moment that determines the cavity tide.

Use the dimensionless moduli k, μ of Eq. (3.1), and put $\lambda_* = k - 2\mu/3$, $f_* = 1 + \lambda_* - \mu$. Substitution of Eq. (5.4) into Eq. (5.2) gives

$$\begin{aligned}
(\lambda_* + 2\mu)(U_0'' + 2U_0'/s - 2U_0/s^2) &= -2f_* s/3, \\
\mu(U_2'' + 2U_2'/s - 8U_2/s^2 + 12V_2/s^2) + (\lambda_* + \mu) D_2' &= 2f_* s/3, \\
\mu(V_2'' + 2V_2'/s - 6V_2/s^2 + 2U_2/s^2) + (\lambda_* + \mu) D_2/s &= f_* s/3,
\end{aligned} \tag{E.2}$$

where $D_0 = U_0' + 2U_0/s$ and $D_2 = U_2' + 2U_2/s - 6V_2/s$. At both $s = 1, \beta$, the three traction conditions are

$$\lambda_* D_0 + 2\mu U_0' = -\lambda_* s^2/3, \qquad \lambda_* D_2 + 2\mu U_2' = \lambda_* s^2/3, \qquad V_2' + (U_2 - V_2)/s = 0. \tag{E.3}$$

The radial equations have the general solution

$$\begin{aligned}
U_0 &= h_0 s + j_0/s^2 - \frac{f_* s^3}{15(\lambda_* + 2\mu)}, \\
U_2 &= 2a_2 s + \frac{6\lambda_* b_2 + f_*/3}{5\lambda_* + 7\mu} s^3 + \frac{(3\lambda_* + 5\mu) c_2}{\mu s^2} - \frac{3d_2}{s^4}, \\
V_2 &= a_2 s + b_2 s^3 + c_2/s^2 + d_2/s^4.
\end{aligned} \tag{E.4}$$

The homogeneous radial powers are s, s^{-2} in the monopole and s, s^3, s^{-2}, s^{-4} in the quadrupole; a cubic particular solution accounts for the source. Positivity of k, μ keeps the displayed denominators nonzero. The six traction conditions fix the six integration constants: the homogeneous problem has zero elastic energy and hence only a rigid displacement, absent in these monopole and spheroidal quadrupole sectors. For $(\beta, k, \mu) = (2, 1/5, 1/10)$ this gives

$$\begin{aligned} U_0 &= \frac{5053}{3150}s + \frac{326}{175s^2} - \frac{31}{150}s^3, \\ U_2 &= 2a_2 s + \left(\frac{24b_2}{41} + \frac{31}{123}\right)s^3 + \frac{9c_2}{s^2} - \frac{3d_2}{s^4}, \\ V_2 &= a_2 s + b_2 s^3 + \frac{c_2}{s^2} + \frac{d_2}{s^4}, \\ (a_2, b_2, c_2, d_2) &= \frac{(-11413363,\ 2029314,\ -3398520,\ -4645536)}{3202122}. \end{aligned} \tag{E.5}$$

At $s = 1, 2$, with $D_2 = U_2' + 2U_2/s - 6V_2/s$, the quadrupole tractions reduce to $2D_2/15 + U_2'/5 = 2s^2/45$ and $V_2' + (U_2 - V_2)/s = 0$. With $D_0 = U_0' + 2U_0/s$, the monopole traction is $2D_0/15 + U_0'/5 = -2s^2/45$ at both faces.

In particular, $U_2(1) = -6246978/533687$ and $U_2(2) = -6078274/533687$. Direct integration of D_2/s gives $\mathcal{C} = 125979/152482$. Thus the face and bulk terms of Eq. (5.11) are $\frac{2}{3}[\mathcal{F}(1) - \mathcal{F}(2)] = 458263/76241$ and $-(1+3k)\mathcal{C} = -503916/381205$, respectively. The transport, pressure, and motion terms in Eq. (5.9) evaluate to

$$\begin{aligned} I_{\text{disp}} &= \frac{790547}{152482}, \qquad I_{\text{stress}} = -\frac{377937}{762410}, \qquad I_{\text{kin}} = -\frac{6}{5}, \\ \mathcal{Q}_c &= \frac{4\pi}{5}(I_{\text{disp}} + I_{\text{stress}} + I_{\text{kin}}) = \frac{5319812\pi}{1906025} > 0. \end{aligned} \tag{E.6}$$

These fractions specify the solution of this particular boundary problem. The physical conclusions use their signs and relative size. In particular, $\mathcal{Q}_c$ is dimensionless; q has dimensions of inverse length squared.

For the contrasting material $k = \mu = 2/5$ and $\beta = 2$, the same six traction equations instead give

$$\begin{aligned} (h_0, j_0) &= \left(\frac{1891}{4410}, \frac{61}{245}\right), \\ (a_2, b_2, c_2, d_2) &= \left(-\frac{1271651}{2185071}, \frac{303495}{2913428}, -\frac{193040}{728357}, -\frac{150784}{728357}\right). \end{aligned} \tag{E.7}$$

Substitute these constants in Eq. (E.4), with $\lambda_* = 2/15$ and $f_* = 11/15$. The face flattenings and volume moment are

$$\mathcal{F}(1) = \frac{1272675}{416204}, \qquad \mathcal{F}(2) = \frac{150630}{104051}, \qquad \mathcal{C} = \frac{24495}{104051}. \tag{E.8}$$

Both flattenings are positive. The inverse moment, however, is

$$\mathcal{Q}_c = \frac{4\pi}{5}\left(\frac{223385}{208102} - \frac{53889}{104051} - \frac{9}{10}\right) = -\frac{716848\pi}{2601275} < 0. \tag{E.9}$$

Thus the sign reversal follows from the same boundary problem and source moment as the original example. The uniform constitutive box in Appendix B applies only to the original moduli; for this second choice, causality and DEC follow in a sufficiently small neighborhood of relaxation from their strict relaxed inequalities.

The monopole gives the central clock coefficient without any additional field equation. Write $c_0 = f_*/[15(\lambda_* + 2\mu)]$ in Eq. (E.4); then $U_0 = h_0 s + j_0/s^2 - c_0 s^3$ and $D_0 = 3h_0 - 5c_0 s^2$. Termwise integration in Eq. (5.13) gives

$$\frac{\mathcal{Z}_c}{4\pi} = \frac{(1+9k)h_0}{2}(\beta^2 - 1) + j_0(1 - \beta^{-1}) - \frac{(1+15k)c_0 + 1 - k}{4}(\beta^4 - 1). \tag{E.10}$$

For the first material, substitute Eq. (E.5) and $c_0 = 31/150$ to obtain $\mathcal{Z}_c/(4\pi) = 4118/2625$. For the second material, $c_0 = 11/210$; inserting Eq. (E.7) gives $\mathcal{Z}_c/(4\pi) = -15929/29400$.

F Rotational first law

Consider a regular, horizon-free member of the rigidly rotating branch. Keep the material annulus, reference metric and constitutive law fixed. The first law relates the change in mass to the change in angular momentum between neighboring equilibria. Its derivation must include the moving faces: although they do no traction work, their displacement changes the domain occupied by the material. For a variation $h_{\mu\nu} = \delta g_{\mu\nu}$, $h = g^{\mu\nu}h_{\mu\nu}$, the Einstein–Hilbert and material actions have boundary potentials

$$\theta^\mu_g = \frac{\sqrt{-g}}{16\pi G}(\nabla_\nu h^{\mu\nu} - \nabla^\mu h), \qquad \theta^\mu_m = -2\sqrt{-g}\,\rho_{H^{AB}}\nabla^\mu X^B \delta X^A. \tag{F.1}$$

Here the indices on $h_{\mu\nu}$ are raised with the background metric. At a moving face with displacement ξ, fixed labels give $\delta X^A = -\xi^\nu \partial_\nu X^A$. The stress identity

$$2\rho_{H^{AB}}\nabla^\mu X^B \partial_\nu X^A = T^\mu{}_\nu + \rho\delta^\mu{}_\nu$$

then yields

$$n_\mu \theta^\mu_m - \sqrt{-g}\rho\, n\cdot\xi = \sqrt{-g}\, n_\mu T^\mu{}_\nu \xi^\nu = 0. \tag{F.2}$$

The second term is the variation of the material domain. Zero traction removes its work at each face; Darmois matching cancels the gravitational interface terms. The regular cavity supplies no inner boundary charge.

Antisymmetrizing the variation of the potentials gives the symplectic form $\mathcal{W}_\Sigma$. Its sign is fixed by $\mathcal{W}(\delta_1, \delta_2) = \int(\delta_1\pi\,\delta_2 q - \delta_2\pi\,\delta_1 q)$ over the material and gravitational

canonical pairs. For variations satisfying the linearized constraints and matching, the Hamiltonian identity [24] for a fixed generator $k = K + \varpi L$ is

$$\delta H_k = \mathcal{W}_\Sigma(\delta, \mathcal{L}_k(g, X)), \qquad H_k = \mathcal{M} - \varpi \mathcal{J}. \tag{F.3}$$

The charge follows from the ADM surface term at infinity; standard asymptotically flat decay and parity conditions make it finite. At a rigidly rotating equilibrium, $\mathcal{L}_k g = 0$ and $kX = 0$, so $\delta\mathcal{M} - \varpi\delta\mathcal{J} = 0$. Applying this identity at each point of the equilibrium branch, with its generator fixed during that variation, gives $d\mathcal{M} = \varpi\, d\mathcal{J}$ at fixed G. Differentiating the smooth odd expansion of $\mathcal{J}$ and integrating this identity gives Eq. (4.9).

G Uniform translation

The velocity of a stationary translation is already restricted at infinity, independently of the constitutive law or internal flow.

Proposition G.1 (Asymptotic translation [25])**.** *Let a spacetime have a Killing field $K = \ell n + Y$ on an asymptotically Euclidean vacuum end of a spacelike hypersurface with future unit normal n. For some $\alpha > 1/2$, assume*

$$\gamma_{ij} - \delta_{ij} = O_2(r^{-\alpha}), \quad K_{ij} = O_1(r^{-1-\alpha}), \quad \ell - a^0 = O_1(r^{-\alpha}), \quad Y^i \to a^i, \tag{G.1}$$

with $\ell, Y \in C^1$ and $a^0 > 0$. Here O_k includes decay of derivatives through order k. If the ADM four-momentum is future timelike, then

$$P^\mu = \frac{P^0}{a^0} a^\mu, \qquad V^i := \frac{a^i}{a^0} = \frac{P^i}{P^0}, \qquad |\mathbf{V}| < 1. \tag{G.2}$$

Proof. The Killing equation gives $\mathcal{L}_Y \gamma_{ij} = 2\ell K_{ij}$ in our sign convention, and the vacuum constraints give $E = S_i = 0$ on the end. Proposition 3.1 of Ref. [25], with the opposite sign for extrinsic curvature, therefore gives $P^\mu \parallel a^\mu$. The remaining conclusions follow from $P^0, a^0 > 0$ and timelikeness of P. □

This established result fixes the asymptotic translation speed even for a circulating shell. It neither forces staticity nor determines an internal clock comparison. The explicit family below realizes it with comoving matter.

Proposition G.2 (Exact coasting family)**.** *Let $(\bar{g}, \bar{X})$ be a spherical elastic equilibrium constructed above, with geometrized ADM mass M. For every $|v| < 1$, set*

$$T = \Gamma_v(t - vx), \quad X = \Gamma_v(x - vt), \quad Y = y, \quad Z = z, \qquad \Gamma_v = (1 - v^2)^{-1/2}. \tag{G.3}$$

Pulling back both $\bar{g}$ and $\bar{X}$ gives an exact Einstein–elastic shell stationary under $K = \Gamma_v(\partial_t + v\partial_x)$, with a flat vacuum cavity, zero surface traction, and the same material energy-condition and characteristic bounds. In the corresponding asymptotic inertial frame,

$$P^\mu = (\Gamma_v M, \Gamma_v M v, 0, 0), \qquad (P^0)^2 - |\mathbf{P}|^2 = M^2. \tag{G.4}$$

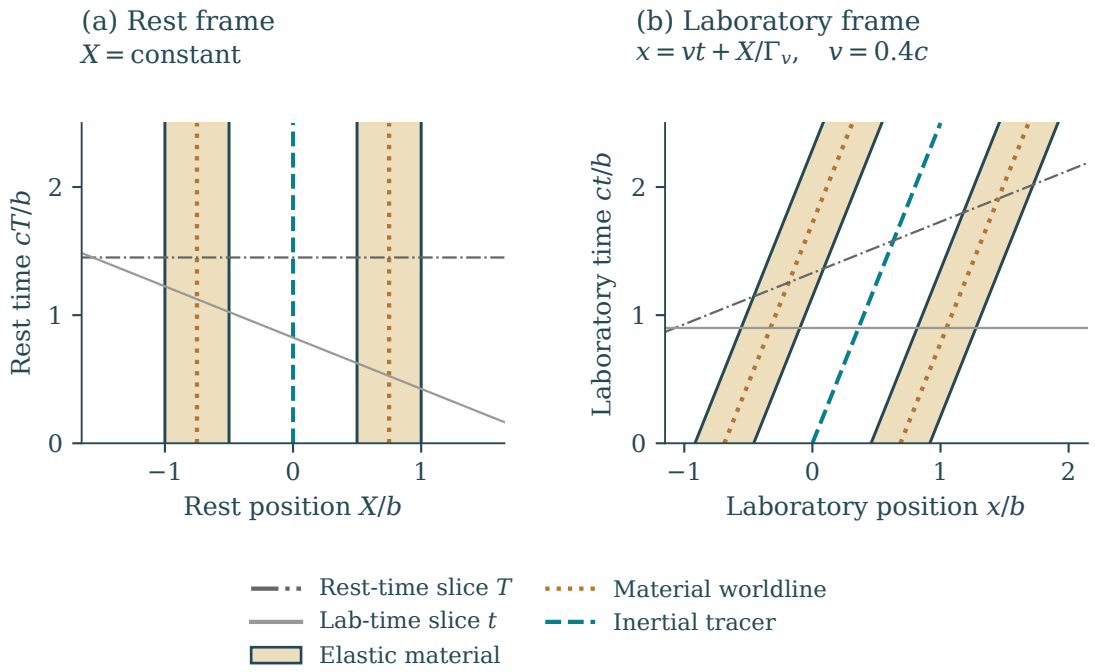

Figure 3. The same shell at rest (a) and coasting (b), in coordinate sections $Y = Z = 0$ and $y = z = 0$ with equal axis scales. The laboratory speed is $v/c = P^x/P^0 = 0.4$. Shading marks material; dotted paths are stress-supported material worldlines, and the dashed path is an inertial cavity tracer. The metric and material undergo the same exact Lorentz pullback, giving uniform motion with conserved ADM momentum. Lines of equal T and equal t show the different simultaneity slices. These are coordinate sections, not measurements of proper shell thickness.

Proof. General covariance preserves the full gravitational and material equations, including the interface conditions, under the same pullback. The map sends K to ∂_T, so the fields are invariant along K and the particles follow it. Flatness, DEC, and local characteristic cones are invariant. The asymptotic transformation is a Lorentz boost [26]; applying it to the rest ADM momentum $(M, 0, 0, 0)$ gives Eq. (G.4). □

The faces in the laboratory coordinates satisfy $\Gamma_v^2(x - vt)^2 + y^2 + z^2 = r(A)^2$ and $r(B)^2$, respectively. Thus this is a uniformly moving material shell with nonzero momentum relative to the chosen asymptotic frame. It is stationary with respect to K, not with respect to laboratory ∂_t. Requiring both time symmetries at nonzero v would make the material support translation invariant along x, which is incompatible with a nonempty compact shell. No small-velocity or weak-field expansion is used in the boost. Figure 3 compares rest and laboratory worldtube sections. The straight material paths in this coordinate diagram retain the static shell's proper acceleration $\sqrt{F}\,\Phi'$; the central cavity tracer is geodesic.

A test passenger initially comoving in the flat cavity remains inertial. Pulling back the perturbations and their Cauchy surfaces carries the radial and odd-parity energy estimates to the corresponding comoving slices. The same pullback preserves the fixed-loop clock experiment of Eq. (4.11); a boost adds no asymmetry to that experiment.

H Gravitational constraints and stress conventions

With $G = 1$, the ADM convention is

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt), \qquad n^a = \alpha^{-1}(\partial_t - \beta^i \partial_i)^a, \tag{H.1}$$

we use $K_{ij} = (-\partial_t \gamma_{ij} + D_i\beta_j + D_j\beta_i)/(2\alpha)$, $E = T_{ab}n^a n^b$, and $S_i = -\gamma_i{}^a T_{ab} n^b$. The constraints are

$$^{(3)}R + K^2 - K_{ij}K^{ij} = 16\pi E, \qquad D_j(K^j{}_i - \delta^j_i K) = 8\pi S_i. \tag{H.2}$$

The spatial Einstein equations, material equations, and free-boundary conditions complete this system.

The timelike material eigenvector makes the elastic stress Type I in every observer frame. For a general source, causal eigenvectors and Jordan structure must be identified at degeneracies [16].

I Coupled radial dynamics

Radial motion changes the strain and gravitational field together. We first eliminate the spherical metric constraints to obtain the oscillation energy, then determine what its nonlinear counterpart controls. Throughout this appendix G is explicit. For a displacement $\xi(t, r)$, the Eulerian variation δ compares fields at fixed radius; the Lagrangian variation $\Delta = \delta + \xi\partial_r$ follows the same material particle. Put $w = \rho + p_r$, $q = p_t - p_r$, and define the material derivatives

$$\mathcal{A} = n_r \partial_{n_r} p_r, \qquad \mathcal{B} = n_t \partial_{n_t} p_r, \qquad \mathcal{D} = n_t \partial_{n_t} p_t, \tag{I.1}$$

where n_t varies both tangential densities together. The linearized Einstein constraints, with no cavity mass perturbation, give

$$\delta m = -4\pi G r^2 w\xi, \qquad \Delta\Lambda = -\Phi'\xi, \qquad \Delta p_r = -\mathcal{A}(\xi' - \Phi'\xi) - \frac{\mathcal{B}}{r}\xi. \tag{I.2}$$

Both moving faces satisfy $\Delta p_r = 0$. Holding their Eulerian pressure perturbation at zero would give a different boundary-value problem.

Set $\zeta = r^2 e^{-\Phi}\xi$ and

$$P = \frac{\mathcal{A}e^{\Lambda+3\Phi}}{r^2}, \qquad W = \frac{we^{3\Lambda+\Phi}}{r^2}, \qquad Y = \frac{2-\mathcal{B}/\mathcal{A}}{r},$$
$$Q = -\frac{e^{\Lambda+3\Phi}}{r^2}\mathcal{V}, \qquad \mathcal{V} = \frac{2q+2\mathcal{D}-\mathcal{B}^2/\mathcal{A}}{r^2} + w\left(\frac{8\pi G p_r}{F} - \Phi'^2 - \frac{4\Phi'}{r}\right) - \frac{4q\Phi'}{r}. \tag{I.3}$$

For a mode $e^{i\sigma t}\zeta(r)$, the complete radial equations reduce to

$$\mathfrak{t} = P(\zeta' - Y\zeta), \qquad \mathfrak{t}' = -Y\mathfrak{t} - (Q+\sigma^2 W)\zeta, \qquad \mathfrak{t}(a) = \mathfrak{t}(b) = 0. \tag{I.4}$$

This is the elastic radial pulsation system [27], here written with two free surfaces. Appendix K derives it with the metric perturbations retained. Spherical symmetry leaves no radiative gravitational mode; those perturbations are reconstructed from the constraints.

Theorem I.1 (Coupled radial stability)**.** *Let a spherical Einstein–elastic equilibrium have a regular vacuum cavity, two free faces at $0 < a < b$, and smooth material coefficients with $F, w, \mathcal{A} > 0$ on $[a, b]$. If*

$$\inf_{[a,b]} e^{2\Phi-2\Lambda}\frac{\mathcal{V}}{w} \geq \gamma > 0, \tag{I.5}$$

then every radial eigenfrequency obeys $\sigma^2 \geq \gamma$. Smooth radial perturbations with the free-traction conditions have a conserved coercive quadratic energy.

Proof. Put $\zeta = e^I y$, where $I(r) = \int_a^r Y(s)\,ds$. Then $-(Pe^{2I}y')' - Qe^{2I}y = \sigma^2 We^{2I}y$, with $y'(a) = y'(b) = 0$. This is a regular Sturm–Liouville problem with positive leading coefficient and weight, so its spectrum is real and discrete. Multiplication by ζ and integration gives

$$\sigma^2\int_a^b W\zeta^2\,dr = \int_a^b \left[P(\zeta'-Y\zeta)^2 - Q\zeta^2\right]dr \geq \gamma\int_a^b W\zeta^2\,dr. \tag{I.6}$$

The same integration for the time-dependent equation conserves

$$E_2 = \frac{1}{2}\int_a^b \left[W\dot{\zeta}^2 + P(\zeta'-Y\zeta)^2 - Q\zeta^2\right]dr. \tag{I.7}$$

Its boundary flux is $[\mathfrak{t}\dot{\zeta}]_a^b = 0$. Positivity controls the velocity and displacement; bounded Y and positive P also control one spatial derivative. □

At zero coupling and relaxation, $\mathcal{V}/w = 12\kappa\mu_0/[m_0(\kappa+4\mu_0/3)r^2] > 0$. Continuity on the compact material interval therefore makes the analytic branch radially stable for sufficiently small $G > 0$. For the numerical example, the sampled minimum of $e^{2\Phi-2\Lambda}\mathcal{V}/w$ is positive, of order $1.8\times10^{-3}L_0^{-2}$, illustrating the sufficient criterion. Nonradial perturbations and rotating backgrounds require separate analyses.

I.1 A nonlinear energy estimate

In spherical comoving coordinates, let V be the proper-time derivative of areal radius and write the geometrized mass as $m = G\mathfrak{m}$. The exact mass constraint defines the energy functional

$$\mathfrak{m}_R = 4\pi r^2 r_R\, \rho\left(\frac{\sqrt{1+V^2-2G\mathfrak{m}/r}}{r_R}, \frac{R}{r}, \frac{R}{r}\right), \quad \mathfrak{m}(A) = 0, \qquad \mathcal{E}_G[r,V] = \mathfrak{m}(B). \tag{I.8}$$

Here $\mathcal{E}_G = M/G$ is the physical ADM energy and M the geometrized ADM mass ($c = 1$). The expression has a smooth $G \to 0$ limit. It is conserved by smooth spherical Einstein–elastic evolution with vacuum cavity and zero outer traction.

Proposition I.2 (Conditional nonlinear radial control)**.** *Fix the material annulus and law (2.6). For an equilibrium satisfying Theorem I.1, there are $c, C > 0$ and a neighborhood $\mathcal{U}$ in $W^{1,\infty} \times L^\infty$ of $(r_*, 0)$ such that*

$$c\|(r-r_*, V)\|_{\mathcal{H}}^2 \le \mathcal{E}_G[r,V] - \mathcal{E}_G[r_*,0] \le C\|(r-r_*, V)\|_{\mathcal{H}}^2, \tag{I.9}$$

where $\|(u,v)\|_{\mathcal{H}}^2 = \int_A^B (R^2 u_R^2 + 2u^2 + R^2 v^2)\, dR$. Every smooth spherical solution remaining in $\mathcal{U}$ therefore obeys $\|(r(t)-r_*, V(t))\|_{\mathcal{H}}^2 \le (C/c)\|(r(0)-r_*, V(0))\|_{\mathcal{H}}^2$.

Proof. The mass functional is stationary at an equilibrium with free faces. Its Hessian, for $\zeta = r^2 e^{-\Phi} u$, is

$$D^2\mathcal{E}_G[(u,v),(u,v)] = 4\pi \int_a^b \left[P(\zeta' - Y\zeta)^2 - Q\zeta^2 + W r^4 v^2\right] dr. \tag{I.10}$$

The derivation is given in Appendix K. The positive radial gap makes this form equivalent to $\|(u,v)\|_{\mathcal{H}}^2$. Smooth coefficients in the mass ODE, with r, r_R and $1 + V^2 - 2G\mathfrak{m}/r$ bounded away from zero, give continuity of its Hessian in this quadratic-form norm on a sufficiently small $\mathcal{U}$. Taylor's formula gives Eq. (I.9); conservation gives the evolution estimate. □

The neighborhood condition is essential. The energy controls one derivative of the displacement in L^2, but not the L^∞ strain or higher derivatives needed for global smooth evolution. This estimate therefore does not exclude shock formation or loss of regularity. General local Einstein–elastic evolution requires compatible initial data at both interfaces, including the higher-order gravitational and traction conditions of Ref. [28]; local existence is not a stability theorem.

Proposition I.3 (Compatible nontrivial radial data)**.** *Let a smooth spherical equilibrium have $r_R, F, w, \partial_{n_r} p_r > 0$ up to both free faces. Every nonzero smooth radial velocity profile v supported away from the faces generates a smooth family of spherical Einstein–elastic initial data with*

$$V_\epsilon = \epsilon v, \qquad r_\epsilon = r_* + O(\epsilon^2), \qquad M_\epsilon = M_* + 2\pi G\epsilon^2 \int_a^b W r^4 v^2 \, dr + o(\epsilon^2). \tag{I.11}$$

The data have the same material content, satisfy both Einstein constraints, and agree with local static solutions near both interfaces. Their physical traction and gravitational matching conditions are therefore compatible to every order. Open constitutive and causal inequalities persist for small $|\epsilon|$.

Proof. Appendix K constructs the data by matching a static outer collar through a scalar mass equation. Its derivative in the unknown outer mass is -1 at equilibrium. The implicit function theorem and the mass Hessian give (I.11). The explicitly reconstructed three-metric and extrinsic curvature satisfy both constraints. □

These data supply examples for the following estimate. Static collars are needed only to construct compatible initial data; the evolution may reflect from either free face.

Corollary I.4 (Radial control through boundary reflections)**.** *Fix a sufficiently weakly gravitating equilibrium, an integer $s \geq 4$, and the time normalization $\alpha(B) = 1$. Let $z = (r, n_r, V, m)$. For every finite $T > 0$, there is $\epsilon_T > 0$ such that smooth spherical initial data satisfying both constraints and all interface compatibility conditions, with $\|z(0) - z_*\|_{H^s} \leq \epsilon < \epsilon_T$, have a unique radial evolution on $[0, T]$, smooth up to each face from either side. Throughout this interval,*

$$\|z(t) - z_*\|_{H^s} \leq C_0 \epsilon e^{Ct}. \tag{I.12}$$

The constants depend on the background, s, and a fixed admissible neighborhood, not on ϵ. The data of Proposition I.3 belong to this class.

Proof. Positive $\partial_{n_r} p_r$ allows radial pressure to replace radial density. In material coordinates both boundary conditions become $p_r = 0$. Time differentiation preserves them, so every differentiated acoustic energy has zero boundary flux. The field equations and propagated constraints recover spatial derivatives without loss. Appendix K gives the resulting estimate $|\dot{E}_s| \leq C E_s$, with $E_s \asymp \|z - z_*\|_{H^s}^2$. Local construction, Gronwall, and continuation then prove the statement: choose ϵ_T so that $C_0 \epsilon_T e^{CT}$ stays inside the fixed neighborhood. □

This controls arbitrarily many reflections on a prescribed finite interval. The allowed amplitude depends on T; no fixed nonzero neighborhood is proved to remain smooth for all time. In particular, the estimate gives neither shock exclusion nor a lifetime bound for a specified disturbance.

I.2 Nonlinear compression and wave breaking

In a nonlinear elastic wave, the propagation speed depends on the local state. Neighboring characteristics can approach one another and steepen a compressive profile while its amplitude stays small. A planar limit of the same material law makes this mechanism explicit before we restore self-gravity and the two free faces. At $G = 0$, take material labels $(X(t, x), y, z)$, so that $n_1 = n = \sqrt{X_x^2 - X_t^2}$, $n_2 = n_3 = 1$, and $v = -X_t/X_x$. The transverse equations vanish, and the longitudinal law is

$$\rho_\parallel(n) = m_0 n + \frac{\kappa}{2}(n-1)^2 + \frac{\mu_0}{6}(n^3 - 2n + n^{-1}), \quad p_\parallel = n\rho_\parallel' - \rho_\parallel, \quad c^2(n) = \frac{n\rho_\parallel''}{\rho_\parallel'}. \tag{I.13}$$

Particle and momentum conservation diagonalize as

$$R_\pm = \operatorname{artanh} v \pm \int_1^n \frac{c(s)}{s}\, ds, \qquad (\partial_t + \lambda_\pm \partial_x) R_\pm = 0, \quad \lambda_\pm = \frac{v \pm c}{1 \pm vc}. \tag{I.14}$$

These combinations are constant along their respective acoustic characteristics and are called Riemann invariants [29]. For $R_- = 0$, $v(n) = \tanh \int_1^n c(s)/s\, ds$ and $\lambda_+'(1) = 3c_0(1 - c_0^2)/2 = 1/\sqrt{3}$ for the selected moduli. A smooth density profile $N(\xi)$ close to one, equal to one outside a finite interval, therefore has the exact solution

$$x = \xi + t\lambda_+(N(\xi)), \qquad n = N(\xi), \qquad n_x = \frac{N'(\xi)}{1 + t\lambda_+'(N(\xi))N'(\xi)}. \tag{I.15}$$

If $a_* := \min_\xi[\lambda_+'(N(\xi))N'(\xi)] < 0$, the map remains invertible for $t < -1/a_*$ and its first Jacobian zero produces an unbounded gradient. The state values are transported unchanged. Particle conservation reconstructs X from $X_x = n/\sqrt{1 - v^2}$ and $X_t = -vX_x$ until that time. The following result retains self-gravity and the finite shell's two faces.

Theorem I.5 (Wave breaking inside a finite shell)**.** *Fix $G > 0$ and a smooth horizon-free spherical equilibrium of the selected elastic law whose principal densities lie strictly inside $[0.99, 1.01]^3$. For every state-amplitude tolerance $\delta > 0$ and time $\tau > 0$, there are smooth radial initial data with the same material annulus, satisfying both Einstein constraints and all interface compatibility conditions, whose classical evolution has a finite maximal time $0 < T_* < \tau$. The data are within δ of equilibrium in velocity, principal densities, and metric coefficients in the uniform norm, using material labels to compare the moving domains. Up to T_* the state remains in a fixed admissible neighborhood, but*

$$\limsup_{t \uparrow T_*} \|\partial_r(n_r, v)(t)\|_{L^\infty} = \infty. \tag{I.16}$$

The singularity occurs in the material interior before a signal from the pulse reaches either face. The exterior mass is allowed to change with the data. No smallness of their higher Sobolev norms is asserted.

Proof. Appendix L derives the exact radial characteristic system and removes the interaction term from its differentiated outgoing equation. The resulting weighted gradient satisfies

$$\frac{dy}{dt} = -a_2 y^2 + a_1 y + a_0, \qquad a_2 \geq a_* > 0, \quad |a_1| + |a_0| \leq C, \tag{I.17}$$

along an outgoing acoustic characteristic. The coefficient bounds depend on the state neighborhood, not on its gradients. Choose an interior velocity pulse of amplitude ε and label width ℓ. Solving the polar-slice mass constraint and adjusting a static outer collar makes the data compatible; its adjustment is $O(\varepsilon^2 \ell)$. At a falling edge, $y(0) \leq -c\varepsilon/\ell + O(1)$.

For $K > 1 + 4C/a_*$, $y(0) \leq -K$ implies $y' \leq -a_* y^2/2$ and preserves $y \leq -K$. Hence $-1/y(t) \leq 1/K - a_* t/2$, which precludes smooth continuation to $2/(a_* K)$. The undifferentiated characteristic equations have bounded sources, so their states remain in the chosen neighborhood for a time independent of ℓ. Take ε small and then ℓ small enough that this upper bound precedes τ and the travel time to either face. The local continuation criterion then forces Eq. (I.16). The static collars remain regular on this interval. □

The result applies in particular whenever the spherical example is enclosed in this strain neighborhood. It supplies no numerical lifetime for a specified disturbance. The ideal elastic action has no microscopic length scale, viscosity, or plastic response: arbitrarily short pulses test that continuum model. A physical prediction after wave breaking requires additional material physics.

J Coupled axial energy

Odd-parity perturbations twist the material and couple to gravitational radiation. Their energy must include the field in the cavity and exterior as well as the elastic motion. The axial equations of relativistic elasticity [30] follow from a quadratic action that makes this energy explicit. First use an axisymmetric representative, and put $\mathfrak{f} = r^2 \sin^2\theta$. Write the perturbed metric and material azimuth as

$$g = \widehat{g} + \mathfrak{f}(d\phi + A_a dx^a)^2, \qquad \phi_{\mathrm{mat}} = \phi + \vartheta, \qquad \mathcal{K}_a = \partial_a \vartheta - A_a, \quad \mathcal{F} = dA, \tag{J.1}$$

where $a \in \{t, r, \theta\}$ and $\widehat{g} = -e^{2\Phi} dt^2 + e^{2\Lambda} dr^2 + r^2 d\theta^2$ is the background quotient metric. The infinitesimal axial gauge acts as $(A, \vartheta) \mapsto (A + d\lambda, \vartheta + \lambda)$. Let $w_t = \rho + p_t$, and let c_r, c_θ be the shear speeds for azimuthal polarization and propagation along the orthonormal radial and polar directions. Define $C^{ij} = c_r^2 e_r^i e_r^j + c_\theta^2 e_\theta^i e_\theta^j$ and $\mathcal{S}^{ab} = -u^a u^b + C^{ab}$ in the material. Products containing w_t vanish in vacuum.

The quadratic action, with the background field equations imposed, is

$$S_2 = -\int \sqrt{-g} \left[\frac{\mathfrak{f}}{64\pi G} \mathcal{F}_{ab} \mathcal{F}^{ab} + \frac{w_t \mathfrak{f}}{2} \mathcal{S}^{ab} \mathcal{K}_a \mathcal{K}_b \right] d^4x. \tag{J.2}$$

The first term follows from reduction along the spacelike axial Killing field. For the second, the local material gradient changes by $\delta F^3{}_i = n_t\sqrt{\mathfrak{f}}\,\mathcal{K}_i$, and its time derivative by $n_t\sqrt{\mathfrak{f}}\,\mathcal{K}_0$. The action Hessian and $n_t^2\mathsf{W}_{33} = w_t$ give the displayed coefficient. This includes the background stress as well as the shear modulus. Variation gives

$$\nabla_b(\mathfrak{f}\mathcal{F}^{ab}) = 16\pi G w_t \mathfrak{f}\mathcal{S}^{ab}\mathcal{K}_b, \qquad \nabla_a(w_t\mathfrak{f}\mathcal{S}^{ab}\mathcal{K}_b) = 0. \tag{J.3}$$

These agree with the covariant axial system of Ref. [30].

Proposition J.1 (Nonnegative coupled axial energy)**.** *Consider a static spherical elastic shell satisfying Einstein's equations, with a regular vacuum cavity, a horizon-free asymptotically flat exterior, and $w_t, c_r^2, c_\theta^2 > 0$ throughout the material. Smooth finite-energy axisymmetric axial perturbations satisfying the linearized constraints and vacuum matching have the nonnegative energy*

$$E_{\mathrm{ax}} = \int_\Sigma \alpha \left[\frac{\mathfrak{f}}{32\pi G}\left(|\mathcal{E}|^2 + \tfrac{1}{2}|\mathcal{B}|^2\right) + \frac{w_t\mathfrak{f}}{2}\left(\mathcal{K}_0^2 + C^{ij}\mathcal{K}_i\mathcal{K}_j\right)\right] dV_\gamma, \tag{J.4}$$

where $\alpha = e^\Phi$, $\mathcal{E}_i = \mathcal{F}_{ti}/\alpha$, $\mathcal{B}_{ij} = \mathcal{F}_{ij}$, and $\mathcal{K}_0 = \mathcal{K}_t/\alpha$. It is conserved when the flux at spatial infinity vanishes. If the null radiation limit exists and satisfies the outgoing falloff below, its flux at future null infinity is nonnegative. There is no exponentially growing finite-energy mode with nonzero $(\mathcal{F}, \mathcal{K}|_{\mathrm{material}})$.

Proof. The momenta of A_i and ϑ are $\Pi^i = \sqrt{\gamma}\,\mathfrak{f}\mathcal{E}^i/(16\pi G)$ and $p = \sqrt{\gamma}\,w_t\mathfrak{f}\mathcal{K}_0$. The Legendre transform of Eq. (J.2) is the density in Eq. (J.4), plus $\Pi^i\partial_i A_t + pA_t$. Integration by parts removes the latter terms by the Gauss constraint $\partial_i\Pi^i = p$ and the asymptotically nonrotating boundary gauge. All remaining coefficients are positive away from the regular axis.

At either material face, the torsional traction is $w_t c_r^2\mathcal{K}_{\hat{r}} = 0$. The metric matching conditions give continuous tangential A in a matching gauge and continuous normal flux $\mathfrak{f}n_b\mathcal{F}^{ab}$. Thus the material energy flux vanishes and the gravitational flux cancels across each interface. Regularity removes axis and center terms. The remaining outward flux density is

$$-\alpha^2\left[\frac{\mathfrak{f}}{16\pi G}\mathcal{B}^{ij}\mathcal{E}_j + w_t\mathfrak{f}C^{ij}\mathcal{K}_j\mathcal{K}_0\right] s_i. \tag{J.5}$$

Here s_i is the outward unit conormal in the background spatial metric. Time translation invariance gives the energy balance. At fixed retarded time, impose

$$\mathcal{E}_{\hat{\theta}} = O(r^{-2}), \qquad \mathcal{B}_{\hat{r}\hat{\theta}} = -\mathcal{E}_{\hat{\theta}} + o(r^{-2}),$$

with regular angular profiles and an integrable radiation limit. The material term is absent there, and the limiting gravitational flux is nonnegative. For a complex mode, use the Hermitian extension of the energy; exponential growth of a nonzero

finite-energy field contradicts conservation. Zero energy sets $\mathcal{F} = 0$ and the material $\mathcal{K} = 0$, leaving gauge and constant label rotations. The argument uses no harmonic cutoff. □

Spherical symmetry removes the restriction to axisymmetric perturbations. For normalized scalar harmonics $Y_{\ell m}$ on the unit sphere, define $L_\ell = \ell(\ell+1)$, $\lambda_\ell = L_\ell - 2$, and $S_A^{\ell m} = \epsilon_A{}^B D_B Y_{\ell m}/\sqrt{L_\ell}$ for $\ell \geq 1$. The vector and tensor normalizations are [31]

$$\int_{S^2} |S^{\ell m}|^2 d\Omega = 1, \qquad 2\int_{S^2} |D_{(A} S^{\ell m}_{B)}|^2 d\Omega = \lambda_\ell. \tag{J.6}$$

In Regge–Wheeler gauge, write $\delta g_{aA} = r^2 a_a(t,r) S_A^{\ell m}$ and the angular label displacement as $q(t,r) S^A_{\ell m}$, with $a \in \{t, r\}$; the odd angular metric perturbation vanishes. Put $e = \dot{a}_r - a_t'$. The corresponding modal energy is

$$E_{\ell m} = \int_0^\infty \alpha e^\Lambda r^4 \Bigg\{ \frac{1}{32\pi G}\left[\frac{e^{-2\Lambda}}{\alpha^2}|e|^2 + \frac{\lambda_\ell}{r^2}\left(\frac{|a_t|^2}{\alpha^2} + e^{-2\Lambda}|a_r|^2\right)\right] + \frac{w_t}{2}\left[\frac{|\dot{q} - a_t|^2}{\alpha^2} + c_r^2 e^{-2\Lambda}|q' - a_r|^2 + \frac{c_\theta^2 \lambda_\ell}{r^2}|q|^2\right] \Bigg\} dr. \tag{J.7}$$

The material terms are integrated only over the shell. For $\ell = 1$ the odd tensor harmonic vanishes; the remaining axial gauge is quotiented.

Corollary J.2 (All odd-parity harmonics)**.** *Under the background hypotheses of Proposition J.1, smooth odd-parity solutions with finite* $E_{\mathrm{odd}} = \sum_{\ell \geq 1} \sum_{m=-\ell}^{\ell} E_{\ell m}$ *obey the same energy and flux conclusions, provided the stated boundary and radiation conditions hold with finite total flux. In particular, there is no exponentially growing finite-energy physical odd-parity mode.*

Proof. The unit-sphere identities $D_A S^A = 0$ and $D^B D_B S_A = (1 - L_\ell) S_A$ give Eq. (J.6) by integration by parts, using $\mathrm{Ric}_{AB} = \Omega_{AB}$. For $m = 0$, put $Z = S^\phi$, so that $(A_t, A_r, \vartheta) = (a_t, a_r, q)Z$ and $A_\theta = 0$. Then $\int \sin^2\theta\, |Z|^2 d\Omega = 1$ and $\int \sin^2\theta\, |Z'|^2 d\Omega = \lambda_\ell$; substitution in Eq. (J.4) gives Eq. (J.7). The background action, constraints, and both interfaces are rotation invariant. Orthogonality of vector and tensor harmonics therefore makes the quadratic action diagonal in (ℓ, m) with these same coefficients for every m. This also transfers the modewise flux identity. Since $\lambda_\ell = (\ell - 1)(\ell + 2) \geq 0$, every summand is nonnegative. Conservation passes to the convergent sum, and monotone convergence applies to the nonnegative outgoing flux. A nonzero growing mode would make at least one of these conserved nonnegative modal energies grow. Completeness of the harmonic expansion leaves only the gauge and label zero modes. □

The constitutive bound (3.6) supplies the material hypotheses whenever the spherical equilibrium remains in the strain box. The $\ell = 1$ sector includes changes in

angular momentum and has no radiative gravitational mode; a comparison at fixed angular momentum removes that stationary variation. There is no axial $\ell = 0$ mode. The estimate retains the regular cavity and the unbounded exterior. It does not address even-parity modes, a rotating background, or nonlinear axial evolution, and supplies no positive frequency gap or decay rate.

In the exterior vacuum, the radiative variable obeys the Regge–Wheeler equation. For $\ell \geq 2$, put $\Psi = r^3 e$ in the vacuum part of the modal action. Its two Euler–Lagrange equations give

$$a_t = -\frac{F(r\Psi)'}{\lambda_\ell r^2}, \qquad a_r = -\frac{\dot{\Psi}}{\lambda_\ell F r}, \qquad \ddot{\Psi} - \partial_{r_*}^2 \Psi + V_\ell \Psi = 0, \quad \frac{dr_*}{dr} = F^{-1}, \tag{J.8}$$

where $V_\ell = F[\ell(\ell+1)/r^2 - 6M/r^3]$. Substitution in the energy and one integration by parts give the usual Regge–Wheeler energy, up to a positive normalization, for compact exterior data. Thus the radiative mode has been retained. The reconstruction divides by λ_ℓ and does not apply to the dipole.

K Radial constraints and the constrained mass

All variations in this appendix preserve the material annulus, its reference metric, and its particle content. Primes denote areal-radius derivatives on the background; R remains the particle label. Put $\mathcal{C} = n_r \partial_{n_r} p_t$. Commutation of the mixed derivatives of $\rho(n_r, n_t, n_t)$ gives

$$\mathcal{B} = 2\mathcal{C} - 2q. \tag{K.1}$$

This identity is needed for self-adjoint radial dynamics; independently prescribed anisotropic pressure perturbations need not satisfy it.

In polar areal coordinates, material kinematics gives $\Delta n_r/n_r = -\xi' - \Delta\Lambda$ and $\Delta n_t/n_t = -\xi/r$. Before eliminating the metric, $\delta\Lambda = \delta m/(rF)$. Differentiating the mass constraint and using $p_r' = -w\Phi' + 2q/r$ yields

$$D_m' + \frac{4\pi G r w}{F} D_m = 0, \qquad D_m = \delta m + 4\pi G r^2 w \xi. \tag{K.2}$$

Vacuum matching at the moving inner face gives $\Delta m(a) = 0$. Since $p_r(a) = 0$, this is $D_m(a) = 0$. Hence $D_m = 0$ throughout, which proves Eq. (I.2). It also gives $\Delta m = -4\pi G r^2 p_r \xi$, so $\Delta m(b) = 0$ at the outer face. The exterior mass has no linear perturbation.

The remaining radial Einstein constraint is

$$\delta\Phi' = \frac{4\pi G r}{F}\delta p_r - \frac{4\pi G w}{F}(1 + 2r\Phi')\xi. \tag{K.3}$$

Normalization at infinity fixes its integration constant. Radial momentum conservation, including its inertial term, reads

$$w e^{2\Lambda - 2\Phi}\ddot{\xi} + \delta p_r' + (\delta\rho + \delta p_r)\Phi' + w\delta\Phi' - \frac{2}{r}(\delta p_t - \delta p_r) = 0. \tag{K.4}$$

Let $\Pi = \Delta p_r$. Substituting the constitutive variations, the constraints, and Eq. (K.1) into the last four terms gives exactly

$$\delta\mathcal{F} = \Pi' + (\Lambda' + 2\Phi' + Y)\Pi + \mathcal{V}\xi, \qquad \Pi = -e^{-\Lambda-2\Phi}\mathfrak{t}. \tag{K.5}$$

Here $\mathcal{F} = p_r' + w\Phi' - 2q/r$ is the static force residual. Equation (I.4) follows, with the elastic pulsation coefficients of Ref. [27]; its anisotropy variable is $q/3$ in our notation. The derivation retains both metric perturbations.

For the nonlinear energy, use comoving spherical coordinates. Then $n_r = e^{-\lambda}$ and $e^{-2\lambda}r_R^2 = 1 + V^2 - 2m/r$. The exact Einstein equations give $m_R = 4\pi G r^2 \rho r_R$ and $u(m) = -4\pi G r^2 p_r V$ [27]. They imply Eq. (I.8) and conservation at the free outer surface. The inner vacuum mass remains zero.

To vary the endpoint mass, write its geometrized ODE as $m_R = f(R, r, r_R, V, m)$. At $V = 0$, direct differentiation gives

$$f_{r_R} = -4\pi G r^2 p_r, \qquad f_m = -\frac{4\pi G r r_R w}{F}. \tag{K.6}$$

The adjoint weight solves $\chi_R = -f_m\chi$, $\chi(B) = 1$. At equilibrium it is $\chi = e^{\Phi+\Lambda}$, because $(\Phi + \Lambda)' = 4\pi G r w/F$ and $\Phi(b) + \Lambda(b) = 0$. For a material displacement $u(R)$, integration by parts gives

$$D\mathcal{E}_G[u, 0] = -4\pi[\chi r^2 p_r u]_A^B + 4\pi \int_A^B \chi r^2 r_R \mathcal{F} u \, dR = 0. \tag{K.7}$$

Thus both free-traction conditions are natural boundary conditions of the mass functional.

Differentiate Eq. (K.7) at equilibrium. Terms multiplying p_r at the faces or $\mathcal{F}$ in the bulk vanish. With $\zeta = r^2 e^{-\Phi} u$, the remaining boundary term is $4\pi[\zeta\mathfrak{t}]_a^b$. Equation (K.5) reduces the volume term to $-4\pi \int_a^b \zeta(\mathfrak{t}' + Y\mathfrak{t} + Q\zeta)\, dr$. Its integration by parts cancels that boundary term and leaves $4\pi \int [P(\zeta' - Y\zeta)^2 - Q\zeta^2] dr$. This calculation allows arbitrary endpoint displacements and does not assume the linearized traction conditions on the variations.

The mass ODE is even in V, so mixed displacement–velocity terms vanish at $V = 0$. Its velocity Hessian is $f_{VV} = 4\pi G r^2 r_R w/F$. Multiplication by χ/G and conversion to dr give the kinetic term $4\pi W r^4 v^2$ in Eq. (I.10). At $G = 0$ and relaxation the full Hessian reduces to

$$4\pi \int_A^B R^2 \left[\kappa(u_R + 2u/R)^2 + \frac{4\mu_0}{3}(u_R - u/R)^2 + m_0 v^2\right] dR. \tag{K.8}$$

The elastic part is bounded below by $c_0(u_R^2 + 2u^2/R^2)$ for any $0 < c_0 \le \min(3\kappa, 2\mu_0)$.

Finally, choose a convex $W^{1,\infty} \times L^\infty$ neighborhood on which r, r_R and $1 + V^2 - 2G\mathfrak{m}/r$ have positive lower bounds. The mass ODE and its first two variational

equations then have uniformly bounded smooth coefficients on the compact material interval. Gronwall's inequality bounds the first mass variation in L^∞ by the $H^1 \times L^2$ variation norm. The second variational equation contains only products of these first variations and of (u, u_R, v). Cauchy–Schwarz therefore bounds its endpoint by the squared norm. The same estimates for differences of coefficients give continuity of the Hessian in quadratic-form norm as the base state varies in this neighborhood. The equilibrium Hessian is coercive by Eq. (I.5); shrinking the neighborhood preserves coercivity. Taylor's integral remainder proves Eq. (I.9). This argument supplies no estimate of higher derivatives of an evolving solution.

K.1 Construction of compatible radial initial data

Write $\Gamma^2 = 1 + V^2 - 2m/r$. On an initial material slice set

$$\gamma = \frac{r_R^2}{\Gamma^2} dR^2 + r^2 d\Omega^2, \qquad K^R{}_R = -\frac{V_R}{r_R}, \qquad K^\theta{}_\theta = K^\phi{}_\phi = -\frac{V}{r}. \tag{K.9}$$

The normal is the material velocity, so the momentum density vanishes. Direct calculation gives

$$D_j(K^j{}_i - \delta^j_i K) = 0, \qquad {}^{(3)}R + K^2 - K_{ij}K^{ij} = \frac{4m_R}{r^2 r_R}. \tag{K.10}$$

Thus the mass ODE (I.8), with $n_r = \Gamma/r_R$ and $n_t = R/r$, solves the Hamiltonian constraint as well.

Fix the equilibrium outer radius b and let M vary near M_*. Solve the static material ODE backward on an outer collar, with terminal data $r(B) = b$, $m(B) = \mathsf{M}$, and $p_r(B) = 0$. Positive $\partial_{n_r} p_r$ determines the terminal radial density and guarantees a smooth family of collar solutions $r^{\mathsf{M}}_{\mathrm{out}}$. Choose a smooth cutoff χ supported in that collar and equal to one on a smaller collar adjoining B, with support disjoint from that of v. Define

$$r_{\mathsf{M}} = r_* + \chi(r^{\mathsf{M}}_{\mathrm{out}} - r_*), \qquad H(\mathsf{M}, \epsilon) = G\mathcal{E}_G[r_{\mathsf{M}}, \epsilon v] - \mathsf{M}. \tag{K.11}$$

At $(M_*, 0)$, $H = 0$, $H_{\mathsf{M}} = -1$, and $H_\epsilon = 0$: the derivatives of $\mathcal{E}_G$ vanish at the free equilibrium for every displacement and velocity. Hence $H(\mathsf{M}(\epsilon), \epsilon) = 0$ has a unique local smooth solution with $\mathsf{M}'(0) = 0$. The second derivative is the kinetic Hessian in (I.10), which proves (I.11).

Integrate the mass constraint from the inner face. Near that face its data are the original static solution. On the outer collar, both this mass function and the local static one obey the same first-order ODE with the same terminal mass, so uniqueness makes them equal there. The reconstructed initial data are therefore static near each face and match flat and Schwarzschild vacuum, respectively. Local static extensions give all time derivatives required for interface compatibility; lapse normalizations on the two collars may differ by constant time rescalings. An initial harmonic gauge

can be chosen for these geometrically compatible data. This construction does not hold the exterior mass fixed while changing the interior energy.

The exact radial evolution also displays the missing derivative control. With $D_t = \alpha^{-1}\partial_t$, $\nu = n_r$, $\eta = R/r$, and $q = p_t - p_r$, its comoving equations are

$$\begin{aligned} D_t r = V, \qquad D_t \nu = -\frac{\nu^2}{\Gamma} V_R, \qquad D_t m = -4\pi G r^2 p_r V, \\ D_t V = -\frac{\Gamma\nu}{w}(p_r)_R + \frac{2\Gamma^2 q}{wr} - \frac{m}{r^2} - 4\pi G r p_r, \\ (\log\alpha)_R = -\frac{(p_r)_R}{w} + \frac{2q r_R}{wr}. \end{aligned} \tag{K.12}$$

The identity $r_R = \Gamma/\nu$ and the mass constraint are propagated. Direct spacetime curvature gives $G^R{}_R = -2D_tV/r - 2m/r^3 + 2\Gamma\nu(\log\alpha)_R/r$; the mixed Einstein equation gives $D_t\nu = -\nu V_R/r_R$. These identities recover the displayed evolution from the radial Einstein equation and stress conservation. With $r_R > 0$, the angular equation follows from the radial contracted Bianchi identity. Thus no independent angular Einstein equation is omitted. For the pair (ν, V) the principal matrix and a symmetrizer are

$$A = \alpha \begin{pmatrix} 0 & \nu^2/\Gamma \\ \Gamma\nu\partial_\nu p_r/w & 0 \end{pmatrix}, \qquad S = \operatorname{diag}\left(\frac{\Gamma^2\partial_\nu p_r}{\nu w}, 1\right). \tag{K.13}$$

Both S and the squared coordinate characteristic speed $\alpha^2\nu^3\partial_\nu p_r/w$ are positive on the admissible branch. The local proper radial distance is $d\ell = dR/\nu$ and proper time is $d\tau = \alpha\, dt$, so the physical squared speed is $\nu\partial_\nu p_r/w$, in agreement with the material acoustic test. Differentiating the traction condition once gives $\mathcal{A}V_R/r_R + \mathcal{B}V/r = 0$ at each face. Higher derivatives of this condition are supplied by the static collars at the initial time. A nonlinear continuation estimate must control the coefficients and differentiated equations in (K.12); the conserved endpoint mass alone does not provide that estimate.

For the boundary estimate, first eliminate r_R and η_R from the force equation using $r_R = \Gamma/\nu$ and $\eta_R = 1/r - \eta\Gamma/(\nu r)$. The lapse introduces no spatial derivative loss. Indeed, define $h(\nu, \eta) = \int_1^\nu (\partial_a p_r)/(\rho + p_r)\, da$. Then

$$(\log\alpha + h)_R = \left(h_\eta - \frac{\partial_\eta p_r}{w}\right)\left(\frac{1}{r} - \frac{\eta\Gamma}{\nu r}\right) + \frac{2q\Gamma}{\nu w r}. \tag{K.14}$$

The right side contains only the state. Integration from B, with $\alpha(B) = 1$, makes $z \mapsto \alpha[z]$ locally Lipschitz in H^s on an admissible neighborhood, including its boundary trace.

The pressure formulation below preserves the constraints, including off-constraint errors. Define $C_1 = r_R - \Gamma/\nu$ and $C_2 = m_R - 4\pi G r^2 \rho r_R$. Direct differentiation, using

the integrated lapse and $[D_t, \partial_R] = (\log\alpha)_R D_t$, gives

$$\begin{aligned} D_t C_1 &= \frac{\eta V}{r}\left(h_\eta - \frac{\partial_\eta p_r}{w}\right) C_1, \\ D_t C_2 &= 4\pi G r^2 \left[\frac{w\nu}{\Gamma} V_R + \frac{V}{r}(2q + \eta\partial_\eta p_r - \eta w h_\eta)\right] C_1. \end{aligned} \tag{K.15}$$

These homogeneous equations keep initially vanishing constraints zero.

K.2 An estimate that includes reflections

Use $p = p_r$ as a variable, writing $\nu = N(p,\eta)$ by $\partial_\nu p_r > 0$. Subscripts t and R on p in this subsection denote derivatives. Put

$$a_p = \frac{\nu^2 \partial_\nu p_r}{\Gamma}, \qquad b_p = \frac{\Gamma\nu}{w}, \qquad f_p = -\frac{\eta\partial_\eta p_r}{r} V, \qquad f_V = \frac{2\Gamma^2 q}{wr} - \frac{m}{r^2} - 4\pi G r p. \tag{K.16}$$

Here the material derivative $\partial_\eta p_r$ is taken at fixed ν. The exact evolution becomes

$$\begin{aligned} &p_t + \alpha a_p V_R = \alpha f_p, \quad V_t + \alpha b_p p_R = \alpha f_V, \\ &r_t = \alpha V, \qquad\qquad m_t = -4\pi G \alpha r^2 p V, \qquad p(t,A) = p(t,B) = 0. \end{aligned} \tag{K.17}$$

The acoustic symmetrizer is $\mathrm{diag}(a_p^{-1}, b_p^{-1})$; its flux is $\alpha P V$ for pressure and velocity variations (P, V). There is one incoming characteristic at each face, with speeds $\pm\alpha\sqrt{a_p b_p}$. The condition $P = 0$ relates the two characteristic amplitudes by reflection and has zero energy flux. No boundary condition is needed for the two ordinary differential equations.

Let $P_j = \partial_t^j (p - p_*)$ and $V_j = \partial_t^j V$. Every P_j vanishes at both faces. Multiplication of the time-differentiated acoustic equations by $(P_j/a_p, V_j/b_p)$ therefore leaves no boundary term, at any order. Take

$$E_s = \frac{1}{2}\sum_{j=0}^{s} \int_A^B \left(\frac{P_j^2}{a_p} + \frac{V_j^2}{b_p}\right) dR + \frac{1}{2}\|(r - r_*, m - m_*)\|_{L^2}^2. \tag{K.18}$$

On a bounded admissible H^s neighborhood this energy is equivalent to $\|z - z_*\|_{H^s}^2$. To see why no derivatives are missing, use

$$\begin{aligned} V_R &= \frac{f_p - p_t/\alpha}{a_p}, \quad p_R = \frac{f_V - V_t/\alpha}{b_p}, \\ r_R &= \frac{\Gamma}{N(p,\eta)}, \qquad m_R = 4\pi G r^2 \rho \frac{\Gamma}{N(p,\eta)}. \end{aligned} \tag{K.19}$$

After subtracting the static equations, induction recovers each mixed space–time derivative of total order at most s from the pure time energies. The last two equations recover one spatial derivative of r, m from lower-order state derivatives. Conversely, the evolution recovers pure time derivatives from the spatial H^s norm.

The lapse boundary trace needs attention in this induction. At B, $p = 0$ implies $\nu = N(0, B/r(B))$. Hence the endpoint value of h in Eq. (K.14) is a smooth function of $r(B)$ alone. Since $r_t(B) = V(B)$, its jth time derivative requires at most the trace of $\partial_t^{j-1} V$. Its H^1 norm is controlled by Eq. (K.19) at order j. The integral part of the lapse is bounded in L^2 by its integrand's L^2 norm. Thus $\partial_t^j \alpha$ is controlled at the same order, without a trace of $\partial_t^j p$ or $\partial_t^j V$.

The one-dimensional product estimates now bound every commutator. The top terms have the form $\partial_t^j a_p \, V_R$, $\partial_t^j b_p \, p_R$, or $\partial_t^j \alpha$ times a first state derivative; that first derivative is bounded in L^∞ for $s \geq 4$. The remaining products have lower total order and the same bound. The equilibrium residual vanishes, so all forcing terms in the difference equations contain a perturbation. Differentiation of the weights and of the last two terms in (K.18) therefore gives

$$|\dot{E}_s| \leq C E_s. \tag{K.20}$$

The constants are uniform in a fixed bounded neighborhood with $r, \nu, \Gamma, w, \partial_\nu p_r$ bounded away from zero.

The construction must also estimate fields that do not yet satisfy the constraints. For this purpose replace the last term of Eq. (K.18) by $\frac{1}{2}\|(r - r_*, m - m_*)\|_{H^s}^2$, and call the resulting energy $\widetilde{E}_s$. The first two equations of Eq. (K.19) recover derivatives of p, V without using either constraint. Together with the retained H^s norms of r, m, they give $\widetilde{E}_s \asymp \|(p - p_*, V, r - r_*, m - m_*)\|_{H^s}^2$. The integrated lapse is a smooth state functional on this space. In particular, its endpoint value involves $r(B)$ through $p(B) = 0$; its differentiated traces are controlled by the acoustic recovery equations as above. The H^s product estimate applied to $r_t = \alpha V$ and $m_t = -4\pi G \alpha r^2 p V$ bounds the derivative of the added norms by $C\widetilde{E}_s$. Thus the energy estimate closes without imposing constraints on intermediate fields.

Freeze the coefficients in (K.17), retain the homogeneous pressure conditions, and solve the acoustic characteristics and the r, m equations with compatible initial time jets. The preceding estimate bounds the iterates in a fixed H^s neighborhood. The difference system has the same zero pressure trace; the state-to-lapse map is locally Lipschitz in H^{s-1}, and the product estimate at that order gives contraction for sufficiently short time. This constructs the radial solution and permits continuation while its H^s norm stays bounded and its state remains admissible. The assumed interface compatibility removes corner terms. Only after taking the limit do Eqs. (K.15) propagate the initially vanishing constraints. Their use then reduces $\widetilde{E}_s$ to the equivalent constrained energy E_s. At either face $p = 0$ keeps the vacuum mass constant; the induced metric and extrinsic curvature match the regular cavity and the Schwarzschild exterior at the moving face. The initial geometric compatibility assumed in Corollary I.4 is the one in Ref. [28]. The argument does not require static collars after the initial slice. Gronwall and Sobolev embedding close the bootstrap on any fixed finite interval at sufficiently small amplitude.

L Interior wave breaking with self-gravity

L.1 A radial system with bounded sources

Use polar areal coordinates while $F = 1 - 2m/r > 0$,

$$ds^2 = -\alpha^2 dt^2 + a^2 dr^2 + r^2 d\Omega^2, \qquad a = F^{-1/2}, \qquad \alpha \to 1 \quad (r \to \infty). \tag{L.1}$$

Here $a(t, r)$ is a metric coefficient, not the inner face radius. Let $v = \tanh \vartheta$ be the radial velocity measured by the slice normal, $\gamma_v = (1 - v^2)^{-1/2}$, $\nu = n_r$, $\eta = R/r$, $p = p_r$, $q = p_t - p$, and $w = \rho + p$. The normal-frame source components are

$$E = w\gamma_v^2 - p, \qquad j = w\gamma_v^2 v, \qquad P = w\gamma_v^2 v^2 + p. \tag{L.2}$$

The radial Einstein equations and the material-label identity give

$$\begin{aligned} m_r &= 4\pi G r^2 E, & m_t &= -4\pi G r^2 \alpha j/a, & (\log\alpha)_r &= a^2(m/r^2 + 4\pi G r P) =: L, \\ \eta_r &= (a\nu\gamma_v - \eta)/r, & \eta_t &= -\alpha\nu\gamma_v v/r, & a_t/(\alpha a) &= -4\pi G r a j =: B_g. \end{aligned} \tag{L.3}$$

In particular, the first derivatives of m, η and the spatial derivative of the lapse depend on state values only.

Define $D = \alpha^{-1}\partial_t + va^{-1}\partial_r$ and $S = v\alpha^{-1}\partial_t + a^{-1}\partial_r$. Particle conservation and the radial projection of stress conservation become

$$\begin{aligned} D\log\nu + S\vartheta &= F_1 := -vL/a - B_g, \\ D\vartheta + c^2 S\log\nu &= F_2 := -L/a - vB_g + \frac{2q}{war} - \frac{p_\eta}{w}\frac{\nu/\gamma_v - \eta/a}{r}, \qquad c^2 = \frac{\nu p_\nu}{w}. \end{aligned} \tag{L.4}$$

Indeed, $D\eta = -v\eta/(ar)$ and $S\eta = (\nu/\gamma_v - \eta/a)/r$; these remove the tangential-density and spherical-divergence terms from the principal part. Hyperelasticity makes the energy equation a consequence of these kinematics and the stress law. Equations (L.3) and (L.4) retain the full radial Einstein–elastic system, including the tangential stress.

Set

$$I(\nu, \eta) = \int_1^\nu \frac{c(s, \eta)}{s}\, ds, \quad U_\pm = \vartheta \pm I(\nu, \eta), \quad \lambda_\pm = \alpha\ell_\pm, \quad \ell_\pm = \frac{v \pm c}{a(1 \pm vc)}. \tag{L.5}$$

Positive sound speed makes this a smooth change of the two acoustic variables at fixed η. Direct substitution gives

$$(\partial_t + \lambda_\pm\partial_r)U_\pm = \alpha f_\pm, \qquad f_\pm = \frac{F_2 \pm cF_1 \pm I_\eta(D\eta \pm cS\eta)}{1 \pm vc}. \tag{L.6}$$

The sources are smooth functions of (U_+, U_-, η, m, r) on any compact admissible state set bounded away from $r = 0$ and $F = 0$.

The outgoing characteristic speed changes strictly with the outgoing wave amplitude throughout the strain box. This property, called genuine nonlinearity, is what permits compression to steepen. At fixed η, write $\rho_\nu = \partial_\nu \rho$ and similarly for higher derivatives. The explicit law gives

$$\frac{\rho_{\nu\nu}}{m_0} = \frac{6\nu + 12\eta^4 + 2\eta^4/\nu^3}{60} > \frac{1}{4}, \qquad \frac{\nu\rho_{\nu\nu\nu}}{m_0} = \frac{\nu - \eta^4/\nu^3}{10}, \qquad |\nu\rho_{\nu\nu\nu}| < \frac{m_0}{100}. \tag{L.7}$$

For example, bound η^4/ν^3 between $(99/100)^4/(101/100)^3$ and $(101/100)^4/(99/100)^3$; substitution gives both displayed bounds. Since $c^2 = \nu\rho_{\nu\nu}/\rho_\nu \le 2/5$,

$$\mathfrak{g} := 1 - c^2 + \frac{\nu c_\nu}{c} = \frac{3}{2}(1 - c^2) + \frac{\nu\rho_{\nu\nu\nu}}{2\rho_{\nu\nu}} > \frac{22}{25}, \qquad \partial_{U_+}\ell_+ = \frac{(1 - v^2)\mathfrak{g}}{2a(1 + vc)^2} > 0. \tag{L.8}$$

The derivative holds U_-, η, m, r fixed. At $G = 0$, $\nu = \eta = 1$ and $v = 0$, it is $1/2$. This positive coefficient drives the steepening of a falling profile.

L.2 Removing the other characteristic gradient

We give the reduction explicitly because bounded sources by themselves would not justify a scalar blowup comparison. Write $u = U_+$, $w_- = U_-$, $z = (\eta, m, r)$ and $d = \ell_+ - \ell_- > 0$. Equations (L.3) determine smooth functions $g(z, u, w_-) = z_r$ and $b(z, u, w_-) = z_t/\alpha$, with the last components 1 and 0. Thus $\alpha^{-1}(\partial_t + \lambda_+\partial_r)z = b + \ell_+ g =: g_+$. All derivatives below in u, w_-, z are ordinary state derivatives. Choose smooth primitives on a small rectangular state neighborhood,

$$h_{w_-} = \frac{(\ell_+)_{w_-}}{d}, \qquad k_{w_-} = -\frac{e^h (f_+)_{w_-}}{d}, \qquad y = e^h \partial_r u + k. \tag{L.9}$$

Integration in w_- at fixed (u, z) defines h and then k; their integration constants may be zero. Neither depends on the lapse. Put

$$\begin{aligned} B_0 &= (f_+)_u - (\ell_+)_z \cdot g - L\ell_+, & C_0 &= (f_+)_z \cdot g + Lf_+, \\ B_1 &= B_0 + h_u f_+ + h_{w_-} f_- + h_z \cdot g_+, & C_1 &= e^h C_0 + k_u f_+ + k_{w_-} f_- + k_z \cdot g_+. \end{aligned} \tag{L.10}$$

Differentiating Eq. (L.6) now gives exactly

$$(\partial_t + \lambda_+\partial_r)y = -\alpha e^{-h}(\ell_+)_u (y - k)^2 + \alpha B_1 (y - k) + \alpha C_1. \tag{L.11}$$

The u_r equation contains the terms $-\alpha(\ell_+)_{w_-} u_r (w_-)_r$ and $\alpha(f_+)_{w_-}(w_-)_r$. The identity

$$(\partial_t + \lambda_+\partial_r)w_- = \alpha f_- + \alpha d (w_-)_r$$

shows how the two primitives in (L.9) cancel them in turn. No derivative of α in time is taken. This avoids estimating a time derivative of a nonlocal lapse integral by an unbounded strain gradient.

On a compact state neighborhood, the lapse is bounded above and below by integrating L through the finite shell and matching to the exterior. Equation (L.8) therefore bounds the quadratic coefficient away from zero; all other coefficients are bounded independently of spatial gradients. Expansion of Eq. (L.11) gives (I.17). This is the usual characteristic steepening argument [29], with the geometric and elastic source terms retained explicitly.

L.3 Compatible pulses and continuation

A polar initial slice differs from the comoving slice in Eq. (I.8). Prescribe an increasing embedding $r(R)$ and a velocity $v(R)$ supported away from the faces. The polar Hamiltonian constraint is the regular scalar ODE

$$\begin{aligned} m_R &= 4\pi G r^2 r_R \left(\frac{\rho + p}{1 - v^2} - p \right), \qquad m(A) = 0, \\ \nu &= \frac{\sqrt{1 - 2m/r}\sqrt{1 - v^2}}{r_R}, \qquad \eta = R/r. \end{aligned} \tag{L.12}$$

The spatial metric is $a^2 dr^2 + r^2 d\Omega^2$. With our sign convention, $K^r{}_r = 4\pi G r a j$ and $K^\theta{}_\theta = K^\phi{}_\phi = 0$ solve the momentum constraint; $K^2 - K_{ij}K^{ij} = 0$, so the Hamiltonian constraint is precisely (L.12).

Use the same static outer collar $r^{\mathsf{M}}_{\mathrm{out}}$ and cutoff as in (K.11), taking $r = r_* + \chi(r^{\mathsf{M}}_{\mathrm{out}} - r_*)$ and $v(R) = \varepsilon f((R - R_0)/\ell)$, where f is a fixed smooth compact bump with a falling edge. Choose its support disjoint from the collar. At $v = 0$, Eq. (L.12) reduces to the static mass functional. Its first variation at equilibrium vanishes for every embedding variation; hence the derivative of $m(B; \mathsf{M}, v) - \mathsf{M}$ with respect to M is -1. The implicit function theorem supplies the matching mass. The source change is $O(v^2)$ on an interval of length $O(\ell)$, so ordinary ODE dependence gives $\mathsf{M} - M_* = O(\varepsilon^2 \ell)$ uniformly in the pulse width. Equivalently, the endpoint map is smooth in the uniform velocity norm; it uses no derivatives of v. The matching solution is static near both faces by ODE uniqueness, and therefore satisfies the geometric and traction compatibility conditions to every order.

Inside the pulse, $r = r_*$, while m_R and η_R remain bounded. Differentiating the density in (L.12) yields

$$\partial_r U_+ = \frac{\gamma_v^2 (1 - cv)}{r_R}\, v_R + O(1). \tag{L.13}$$

The remainder is bounded independently of ℓ. The positive coefficient of v_R proves the initial compression asserted in the theorem.

For completeness, the continuation argument uses state bounds before it uses derivative bounds. Equations (L.6) along their two characteristics, and the bounded equations for z, keep the values in a fixed compact neighborhood for a short time independent of ℓ. Choose a wider interior interval around the pulse. Its distance from

either face and the bounded light speed α/a give a positive travel time independent of ℓ. The complementary initial regions have uniform smooth bounds; near each face they are exactly static until an incoming signal arrives. The distant pulse can change the lapse there by a spatially constant factor depending on time. This is a time reparametrization of the static collar and produces no material forcing.

Smooth local evolution follows from the radial hyperbolic system and the compatible free-face construction. If the state stays admissible and $\|\partial_r(\nu, v)\|_\infty$ stays bounded on a finite time interval, differentiating the system gives linear estimates for each higher Sobolev norm with coefficients controlled by these first derivatives. The radial constraints recover the metric and label derivatives without loss; the free-face energy has zero traction work as in Appendix K. Thus the solution continues unless a first state derivative becomes unbounded. Apply the Riccati comparison before the uniform state and travel times. Any earlier breakdown has the same continuation obstruction, and the uniformly regular complementary regions place it in the material interior. This proves Theorem I.5. Entropy-admissible weak evolution beyond T_* is a separate problem.

Data availability

The numerical examples use no external observational or experimental data. The constitutive law, field equations, boundary conditions, parameters, and derivations needed to reproduce them are given in the article.

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